The Weight of the Adjoint

Which Symplectic Form a Neural Operator Preserves, and Whether It Survives the Lift

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Abstract

Structure-preserving neural operators are built on a question that is rarely stated explicitly: symplectic *with respect to which inner product*? On a non-uniform grid the discrete L^2 adjoint is $K^* = W^{-1}K^\top W$, where W holds the quadrature weights, and the choice of W selects which symplectic form a gradient shear preserves. We make the choice explicit and measure it. The symplectic defect, computed from the autograd Jacobian, is exact (4×10^{-16}) in the form a shear was built from and $O(1)$ in the others, in 1-D, 2-D and 3-D. Does matching the form to the grid help? On uniform grids the matched form has the lower error in 17 of 22 paired comparisons over five seeds, and none favours the other. On Chebyshev nodes, where the roles reverse, it is better in two of three comparisons and tied in the third; on a mapped grid where neither standard form is right it brings no consistent benefit. Because the weights also set the kernel's quadrature, the experiments cannot separate the form from the quadrature rule. Over 2×10^4-step rollouts, end-to-end symplectic maps stayed bounded in 60 of 63 runs (the exceptions: one mismatched-form model in 3-D), while a lifted variant with a non-symplectic linear lift and projection was bounded in 3 of 12. Bounded is not accurate: an identity map is bounded too, the error is comparable to the signal, and energy is off by over 100%, against 0.03% for the reference solver. A 783-run benchmark places the constructions among standard neural operators, and a worked example shows that a unit Jacobian determinant, which every shear has, does not make an update symplectic.

1 Introduction

Neural operators learn maps between function spaces and are now widely studied as surrogates for parametric PDEs (Kovachki et al., 2023; Li et al., 2021; Lu et al., 2021). A fast-growing branch argues that such operators should respect the *geometric structure* of the dynamics they emulate: symplecticity for Hamiltonian systems, or the metriplectic split when dissipation is present. The argument has a strong pedigree in finite dimensions (Greydanus et al., 2019; Jin et al., 2020; Cranmer et al., 2020; Zhong et al., 2020), and in 2026 several groups carried it into function space (Makara et al., 2026; Sulskis and Ravi, 2026; Zhang et al., 2026; Varghese et al., 2026).

The function-space constructions we reviewed establish structure preservation by construction. The Symplectic Neural Operator (SNO), for instance, defines the symplectic form from a Hilbert inner product and notes that self-adjointness depends on that choice, but its self-adjoint layers are Fourier operators on the torus and its experiments use uniform grids (Makara et al., 2026), where the quadrature weight is a single scalar and the weighted adjoint reduces to the transpose. On a non-uniform grid the choice matters: the weights vary by a factor of order N (Figure 1), the adjoint is $K^* = W^{-1}K^\top W$ rather than $K^\top$, and two implementations that differ in which W they assume

preserve two *different* symplectic forms. We are not aware of a measurement of the symplectic defect of a trained operator against more than one candidate inner product; this is the measurement we make.

This gives an unusually clean experiment. Build both, verify that each is exact in its own form, then ask which one helps. We organise the study around three questions:

RQ1. Does each construction preserve the form it claims, and only that one? (§6.1)

RQ2. Does matching the form to the discretisation of the data improve prediction, on uniform, Chebyshev and mapped grids? (§6.2)

RQ3. Does the guarantee survive the lift, projection and residual update that conventionally wrap the core, and what does end-to-end symplecticity buy over a long rollout? (§6.3)

The mathematics is in §3, kept separate from the experiments that test it; §3.6 states what it predicts for each question. A broader nine-equation benchmark (§7) is secondary: it places the constructions among standard operators, and none of the three questions rests on it. Our contributions:

1. **An instrument** (§3.5). We measure the symplectic defect directly from the autograd Jacobian, against *both* candidate inner products, scaling to 3-D with Hutchinson probes (Hutchinson, 1990; Avron and Toledo, 2011) whose error we measure against the dense value. It shows that a Störmer–Verlet-style update with a unit Jacobian determinant has an $O(1)$ defect in both candidate forms (Appendix A).

2. **A construction** (§3.4, §4). SA-Cheb: exact symplecticity on a non-periodic Chebyshev grid in 1-D, 2-D and 3-D, for any linear kernel and any continuously differentiable activation (Theorem 5). The gradient-shear structure follows Makara et al. (2026); the weighted adjoint on non-periodic nodes is what we add. We describe it, and the non-gradient kernel operator CKINO that serves as a baseline, component by component.

3. **The form result** (RQ1 and RQ2). Each variant is exact in its own form and $O(1)$ in the others. On uniform grids the matched form has the lower error in 17 of 22 paired comparisons over five seeds (16 of them in every seed) and none favours the other form; on Chebyshev nodes the preference reverses in two of three comparisons and ties in the third; on a mapped grid where neither standard form is the quadrature it brings no consistent benefit. The weights also set the quadrature of the kernel integral, so the form cannot be separated from the quadrature rule.

4. **The end-to-end result** (RQ3). Over 2×10^4 steps end-to-end symplectic maps stayed bounded in 60 of 63 runs, the exceptions being the mismatched-form model in 3-D; a lifted variant with a non-symplectic linear lift and projection was bounded in 3 of 12. Controls temper the reading: an identity map is bounded too, an untrained symplectic map diverges, and a 1–10% perturbation of the adjoint is still bounded. Boundedness tracked the spectral radius of the one-step Jacobian. Bounded is not accurate: at that horizon the error is comparable to the signal and energy is not conserved.

5. **A benchmark** (§5, §7): 783 runs over nine equations in 1-D/2-D/3-D at matched parameter count, three seeds, plus an unconstrained-capacity sweep showing the ranking is budget-dependent, and 338 further training runs for the form and long-horizon experiments (two more seeds for the headline pairs, two further grids, long-horizon controls and a multi-resolution test).

2 Related work

Neural operators. The Fourier Neural Operator parameterises the integral kernel by truncated Fourier multipliers (Li et al., 2021), inheriting discretisation invariance and $O(N \log N)$ cost; DeepONet factorises the operator into branch and trunk networks (Lu et al., 2021); the general framework and universality results are due to Kovachki et al. (2023). Tensorised (Kossaifi et al., 2023) and U-Net-augmented (Wen et al., 2022) variants improve parameter efficiency. A practitioner-oriented account of FNO's assumptions is given by Duruisseaux et al. (2025), with reference implementations in Kossaifi et al. (2024). Physics-informed variants add PDE residuals to the loss (Raissi et al., 2019; Li et al., 2024); these are *soft* constraints with no algebraic guarantee.

Structure preservation in finite dimensions. Hamiltonian (Greydanus et al., 2019) and Lagrangian (Cranmer et al., 2020) neural networks learn the generating function; SympNets (Jin et al., 2020) and Symplectic ODE-Net (Zhong et al., 2020) parameterise symplectic maps directly. These act on $\mathbb{R}^{2d}$ at a fixed discretisation and are not operators in the function-space sense.

Structure-preserving neural operators. Makara et al. (2026) introduce the Symplectic Neural Operator (SNO), composing symplectic shear blocks whose update fields are gradients of learned energy functionals, realised with self-adjoint Fourier layers on the torus, and give a modified-Hamiltonian stability result for the learned map under stated assumptions. Sulskis and Ravi (2026) embed the GENERIC degeneracy structure (Grmela and Öttinger, 1997; Öttinger, 2005) into an FNO, extending finite-dimensional bracket learning (Zhang et al., 2022). Zhang et al. (2026) compose structure-preserving blocks on boundary-adapted spectral representations, and Varghese et al. (2026) apply a conditional symplectic operator to optimal control.

Where we differ. The self-adjointness result of SNO is stated on the torus with the Haar probability measure and its experiments use uniform grids (Makara et al., 2026), where $W \propto I$ and the weighted adjoint reduces to the transpose; GENERIC-FNO is built from Fourier multipliers (Sulskis and Ravi, 2026), and PI-SONet acts on latent spaces of control problems (Varghese et al., 2026). LegONet represents fields in boundary-adapted bases and builds its structure operators from the discrete L^2 metric (the mass matrix) of the basis (Zhang et al., 2026), but its blocks are generator-induced vector fields in coefficient space advanced by operator splitting, so the question we study, which weights make a nodal gradient shear symplectic, is not posed there. Our shear field $F = K^*\rho(Kq)$ is the weighted-adjoint analogue of SNO's gradient neural functional, with the quadrature weights of the grid in place of the Haar measure. Makara et al. (2026) state that their formulation applies to strong canonical symplectic Hilbert phase spaces and that weak symplectic forms and general manifolds need additional theory; we do not address those either. We work where the choice of weights is forced, show that it is consequential, and measure rather than assert the outcome. Lie-equivariant lifting (Finzi et al., 2020) is a standard ingredient and not claimed as a contribution.

3 Theory: which symplectic form a gradient shear preserves

This section contains everything the paper claims as mathematics, and nothing that depends on data. It fixes notation and the discrete inner product (§3.1, §3.2), gives the criterion for a shear to preserve a weighted symplectic form (§3.3), constructs a gradient shear that meets the criterion exactly in a chosen form and in no other (§3.4), and defines the measurement used to test this, with

its estimator (§3.5). It closes with the predictions the experiments test, one per research question (§3.6).

3.1 Notation

Table 1 collects every symbol used in the paper. Vectors are columns, $^\top$ is the plain matrix transpose, and * is reserved for the adjoint with respect to a weighted inner product; the distinction between the two is the subject of the paper.

Table 1: Symbols and notation.

symbol	reads as	meaning
Ω_{d}		spatial domain, taken as $[-1, 1]$ in 1-D
d		number of spatial dimensions, $d \in \{1, 2, 3\}$
N		grid index; a 1-D grid carries $N+1$ nodes
$x_j = \cos(j\pi/N)$		Chebyshev–Gauss–Lobatto (CGL) nodes
$w_j > 0$		quadrature weight at node j
$W = \mathrm{diag}(w_j)$		weight matrix; *diagonal and positive*
W_{cheb}, W_{unif}		the two candidate weight matrices: Clenshaw–Curtis weights on CGL nodes, and constant weights (a uniform grid)
$\langle f, g\rangle_W = f^\top W g$	"f dot g in W"	discrete L^2 inner product
K		a linear (kernel-integral) operator
$K^\top$	"K transpose"	matrix transpose
$K^* = W^{-1}K^\top W$	"K adjoint"	adjoint *with respect to* $\langle\cdot,\cdot\rangle_W$
q, p		the two halves of a canonical phase-space state
$z = (q, p)$		a point of phase space $\mathbb{P}$
ω_W	"omega"	symplectic form induced by W, eq. (2)
Ω		matrix of ω_W, eq. (2)
Φ	"Phi"	one shear map of phase space
$D\Phi$		Jacobian of Φ
F, G		the update (shear) fields
$A = DF$		Jacobian of the shear field
R, $\rho = R'$	"rho"	scalar potential and its derivative, applied pointwise
$D = \mathrm{diag}(\rho'(\cdot))$		diagonal matrix of pointwise slopes
$\mathcal{E}$	"script E"	learned energy functional
∇_W	"grad in W"	gradient with respect to $\langle\cdot,\cdot\rangle_W$
φ_r, ψ_r	"phi, psi"	the r-th left/right basis functions of the kernel
M_r		channel-mixing matrix of mode r
R (rank)		Mercer rank; context distinguishes it from the potential R
σ_r	"sigma"	learned positive Mercer scale
C, H		physical and hidden channel counts
$\mathcal{D}_W$	"defect"	relative symplectic defect, eq. (9)
$\|\cdot\|_F$	"Frobenius norm"	$\|X\|_F^2 = \sum_{ij} X_{ij}^2$
$[S, M] = SM - MS$	"commutator"	used in Proposition 7

3.2 The discrete inner product

Let $\Omega_{\mathrm{d}} = [-1, 1]$ be discretised on the $N + 1$ CGL nodes $x_j = \cos(j\pi/N)$. Functions are represented by nodal values, and the physical L^2 product is approximated by Clenshaw–Curtis quadrature

(Clenshaw and Curtis, 1960; Trefethen, 2000),

$$\langle f, g\rangle_W \;=\; \int_{\Omega_{\mathrm{d}}} f\,g\,dx \;\approx\; f^{\top} W g, \qquad W = \mathrm{diag}(w_0, \ldots, w_N), \quad w_j > 0. \tag{1}$$

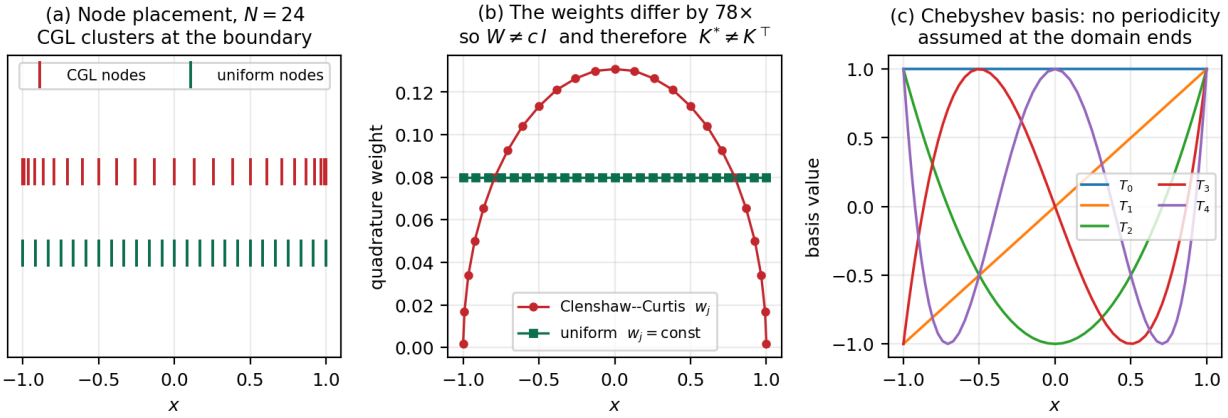


Figure 1: The geometric fact the paper rests on. **(a)** CGL nodes cluster quadratically at the boundary; a uniform grid does not. **(b)** The Clenshaw–Curtis weights therefore vary by 78× across the domain at $N = 24$, so $W \neq cI$ and consequently $K^* \neq K^{\top}$. On a uniform grid the two coincide, which is why a Fourier-parameterised operator using the standard uniform quadrature does not meet the distinction. **(c)** The Chebyshev basis assumes no periodicity at the domain ends.

Phase space is $\mathbb{P} = \mathcal{H}_q \oplus \mathcal{H}_p$ with $\mathcal{H} = (\mathbb{R}^{N+1}, \langle\cdot,\cdot\rangle_W)$, carrying

$$\omega_W(z, z') = \langle q, p'\rangle_W - \langle p, q'\rangle_W = z^{\top}\Omega z', \qquad \Omega = \begin{pmatrix} 0 & W \\ -W & 0 \end{pmatrix}. \tag{2}$$

This is the finite-dimensional shadow of the weak symplectic form on a Hilbert phase space (Weinstein, 1969; Marsden and Ratiu, 1999); the point of what follows is that the discretisation chooses W, and that choice is not innocuous.

Definition 1 (Weighted adjoint)**.** The adjoint of K with respect to (1) is the unique K^* with $\langle Kf, g\rangle_W = \langle f, K^* g\rangle_W$ for all f, g. Expanding, $f^{\top}K^{\top}Wg = f^{\top}WK^*g$, hence

$$\boxed{K^* = W^{-1}K^{\top}W} \tag{3}$$

3.3 When is a shear symplectic?

Proposition 2 (Shear criterion)**.** *Let $\Phi(q, p) = (q,\, p + F(q))$ with $A = DF(q)$. Then*

$$(D\Phi)^{\top}\Omega\,(D\Phi) - \Omega = \begin{pmatrix} WA - A^{\top}W & 0 \\ 0 & 0 \end{pmatrix}, \tag{4}$$

so Φ preserves ω_W iff A is self-adjoint in W, i.e. $WA = A^{\top}W$. The transposed shear behaves analogously.

Proof. $D\Phi = \left(\begin{smallmatrix} I & 0 \\ A & I \end{smallmatrix}\right)$, so $\Omega D\Phi = \left(\begin{smallmatrix} WA & W \\ -W & 0 \end{smallmatrix}\right)$ and $(D\Phi)^{\top}\Omega D\Phi = \left(\begin{smallmatrix} I & A^{\top} \\ 0 & I \end{smallmatrix}\right)\left(\begin{smallmatrix} WA & W \\ -W & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} WA - A^{\top}W & W \\ -W & 0 \end{smallmatrix}\right)$. Subtract Ω. □

Remark 3 (Volume preservation is not symplecticity). $\det D\Phi = 1$ for *every* shear, whatever A. Unit Jacobian determinant is Liouville volume preservation (Arnold, 1989), strictly weaker than preservation of ω_W, which by Proposition 2 additionally requires $WA = A^\top W$. Conflating the two is an easy way for an architecture described as symplectic to fall short of the claim; §6.1 and Appendix A give an instance, a Störmer–Verlet-style block with a unit-triangular Jacobian whose measured defect is $O(1)$ in both candidate forms.

Remark 4 (The question is under-specified). Because (4) depends on W, "is this block symplectic?" has no answer until W is named. Two shears built with the adjoint taken in different W are *both* exactly symplectic, in different forms. This is the observation the rest of the paper develops. (In our implementation W also enters the kernel K itself, as its quadrature; the theorem holds for any K, so this does not affect it, but it matters for what the experiments can separate, §4.2.)

3.4 A gradient shear that is exact in a chosen form

The construction below makes the self-adjointness required by Proposition 2 hold by design. The state q carries c channels on the $(N+1)^d$ grid nodes, $q \in \mathbb{R}^{c\times(N+1)^d}$. Let $K : \mathbb{R}^{c\times(N+1)^d} \to \mathbb{R}^{h\times(N+1)^d}$ be linear, that is, a matrix acting on the flattened channel–node index; the low-rank Chebyshev kernel used in practice is one such matrix (§4.2). The weight acts channel by channel, $W_{\text{in}} = I_c \otimes W$ on the domain and $W_{\text{out}} = I_h \otimes W$ on the range, both diagonal and positive, with $W = \bigotimes_{a=1}^{d} W_a$ in d dimensions. The weighted adjoint is then $K^* = W_{\text{in}}^{-1} K^\top W_{\text{out}}$, which is (3) when $c = h = 1$, and the phase-space form is $\omega_{W_{\text{in}}}$. Let $R \in C^2(\mathbb{R})$ with $\rho = R'$, both applied entrywise, and define the energy functional and its gradient in $\langle\cdot,\cdot\rangle_{W_{\text{in}}}$,

$$\mathcal{E}(q) = \mathbf{1}^\top W_{\text{out}}\, R(Kq) \;\approx\; \sum_{\text{channels}} \int_\Omega R\big((Kq)(x)\big)\, dx, \qquad F(q) \;=\; \nabla_{W_{\text{in}}}\mathcal{E}(q) \;=\; K^*\rho(Kq). \tag{5}$$

The block alternates

$$\Phi^{\uparrow}(q,p) = (q,\; p + F_\theta(q)), \qquad \Phi^{\downarrow}(q,p) = (q + G_\vartheta(p),\; p). \tag{6}$$

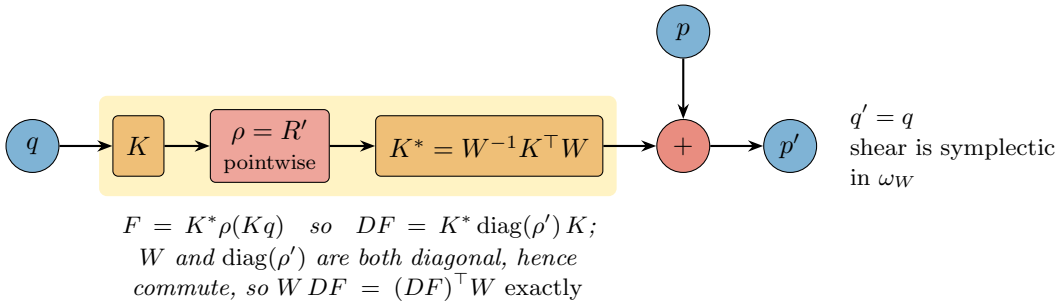


Figure 2: The SA-Cheb gradient shear, drawn for one channel (in general $K^* = W_{\text{in}}^{-1} K^\top W_{\text{out}}$). The design choice is which weights W define the quadrature inside K and the adjoint K^*: the Clenshaw–Curtis weights (W_{cheb}) or constant weights (W_{unif}). Both give an *exactly* symplectic block, in different forms. That choice, not the presence of symplecticity, is what the experiments vary.

Theorem 5 (Exact symplecticity, any dimension)**.** *Let K, W_{in}, W_{out} and R be as above. For any entries of K, any $R \in C^2$ (equivalently $\rho \in C^1$), any resolution N, any channel counts c, h and any spatial dimension $d \in \{1,2,3\}$, the map $\Phi^{\uparrow}$ of (6) satisfies $(\Phi^{\uparrow})^*\omega_{W_{\text{in}}} = \omega_{W_{\text{in}}}$ exactly, and likewise $\Phi^{\downarrow}$; hence so does any finite alternating composition.*

Proof. Write $D = \operatorname{diag}\big(\rho'((Kq)_k)\big)$, diagonal over the channel–node index k of the range, so $A = DF = K^*DK$. Since $(K^*)^\top = W_{\text{out}} K W_{\text{in}}^{-1}$,

$$W_{\text{in}} A = W_{\text{in}}\big(W_{\text{in}}^{-1} K^\top W_{\text{out}}\big) DK = K^\top W_{\text{out}} DK,$$
$$A^\top W_{\text{in}} = K^\top D\, W_{\text{out}} K W_{\text{in}}^{-1} W_{\text{in}} = K^\top D W_{\text{out}} K.$$

W_{out} and D are both diagonal on the range and therefore commute, so $W_{\text{in}} A = A^\top W_{\text{in}}$; apply Proposition 2 with W_{in} in place of W. The dimension d and the channel counts enter only through the sizes of these diagonal matrices. For $\Phi^\downarrow$ exchange the roles of q and p. □

Remark 6 (Regularity of the activation). The proof uses ρ' only through D, so it needs ρ to be differentiable at the points $(Kq)_k$. Our implementation uses $\rho = \text{GELU}$ (Hendrycks and Gimpel, 2016), which is C^∞, so the theorem holds at every state. For a piecewise-C^1 activation such as ReLU, ρ' fails to exist only where some $(Kq)_k$ sits at the kink, a finite union of hyperplanes in q. Away from that set the identity $W_{\text{in}} A = A^\top W_{\text{in}}$ holds as stated, so the shear is symplectic almost everywhere; on it the Jacobian is undefined.

Theorem 5 says the block is symplectic in *the form its adjoint was built from.* It says nothing about any other form, and the next result shows that the omission is not a technicality: it is the whole content of the paper, and it predicts the pattern that RQ1 tests (§6.1).

Proposition 7 (Which form, and only which form)**.** *Fix a positive diagonal W_1 and build the shear field with the adjoint taken in W_1, i.e. $F(q) = K^*_{W_1}\rho(Kq)$ with $K^*_{W_1} = W_1^{-1}K^\top W_1$, so that $A = K^*_{W_1}DK$. Let W_2 be any other positive diagonal weight. Then*

$$W_2 A - A^\top W_2 \;=\; [\,S,\,M\,] \;=\; SM - MS, \qquad S := W_2 W_1^{-1}, \quad M := K^\top W_1 DK, \tag{7}$$

where S is diagonal and M is symmetric. Consequently

$$\big(W_2 A - A^\top W_2\big)_{ij} = (s_i - s_j)\, M_{ij}, \tag{8}$$

and the shear is symplectic in ω_{W_2} for every K and ρ if and only if $W_2 \propto W_1$. We state this for $c = h = 1$; with channels, each W_i becomes $I \otimes W_i$ on the domain and on the range, and the proof is unchanged.

Proof. Since W_1 is symmetric, $(K^*_{W_1})^\top = (W_1^{-1}K^\top W_1)^\top = W_1 K W_1^{-1}$. Diagonal matrices commute, so $W_1 D = D W_1$ and $W_1^{-1}W_2 = W_2 W_1^{-1} = S$. Then

$$W_2 A = W_2 W_1^{-1} K^\top W_1 DK = S\,\big(K^\top W_1 DK\big) = SM,$$

$$A^\top W_2 = K^\top D\,(K^*_{W_1})^\top W_2 = K^\top D\, W_1 K W_1^{-1} W_2 = \big(K^\top W_1 DK\big) S = MS,$$

which gives (7). M is symmetric because $M^\top = K^\top (W_1 D)^\top K = K^\top D W_1 K = K^\top W_1 DK = M$. Writing $S = \operatorname{diag}(s_i)$ gives $(SM - MS)_{ij} = s_i M_{ij} - M_{ij} s_j$, which is (8).

For the equivalence: if $W_2 = cW_1$ then $S = cI$, every $s_i - s_j$ vanishes and so does the defect, recovering Theorem 5 at $c = 1$. Conversely, suppose $W_2 \not\propto W_1$, so there are indices $i \neq j$ with $s_i \neq s_j$. Choose $D = I$ (take $\rho' \equiv 1$) and $K = e_i e_i^\top + e_j e_j^\top$; then $M = K^\top W_1 K$ has $M_{ij} = 0$, so pick instead $K = (e_i + e_j)e_i^\top + (e_i + e_j)e_j^\top$, for which $M_{ij} = (w_{1,i} + w_{1,j}) \neq 0$. By (8) the (i,j) entry of the defect is $(s_i - s_j)M_{ij} \neq 0$, so the shear is not symplectic in ω_{W_2} for that parameter choice. □

Remark 8 (The defect scales with the spread of W_2/W_1). Equation (8) says the mismatch is governed by the *spread* of the ratio $s_i = w_{2,i}/w_{1,i}$: for a fixed M every entry of the defect scales with a difference $s_i - s_j$, so a wider dynamic range in $W_2 W_1^{-1}$ should produce a larger defect. This is a heuristic rather than a bound, because the normalisation of the defect in (9) below also depends on W_2, but it is falsifiable. For W_{cheb} against W_{unif} the per-axis ratio spans a factor of $\approx \pi N$, since the Clenshaw–Curtis weights fall from $\approx \pi/N$ in the interior to $O(N^{-2})$ at the ends, and the tensor-product weight $\bigotimes_a W_a$ compounds this to $(\pi N)^d$. The off-diagonal defect should therefore grow with dimension; RQ1 tests this (§6.1).

Low-rank realisation. For the low-rank kernel and its weighted adjoint (Appendix B),

$$Kv = \sum_r \varphi_r M_r \langle \psi_r, v \rangle_W, \qquad K^* g = \sum_r \psi_r M_r^\top \langle \varphi_r, g \rangle_W :$$

swap $\varphi \leftrightarrow \psi$ *and transpose* M. No inverse of W is ever formed and the cost stays $O(RN^d)$.

3.5 Measuring the defect

Proposition 2 turns the question into a number. For a shear field F we form $A = DF$ by reverse-mode differentiation and report

$$\mathcal{D}_W(F) \;=\; \frac{\|WA - A^\top W\|_F}{\|WA\|_F}, \tag{9}$$

zero iff the block preserves ω_W. Crucially we evaluate (9) for *both* $W \in \{W_{\text{cheb}}, W_{\text{unif}}\}$: reporting one column can hide the matched form altogether, since a map that is exactly symplectic in W_{unif} has an $O(1)$ defect when scored only against W_{cheb}.

Scaling to 3-D. A dense Jacobian costs one backward pass per degree of freedom: 65 passes for a 1-D grid at $N = 64$, but $17^3 = 4913$ at $N = 16$ in 3-D, for every model, form and seed. Proposition 9 replaces it with a probe estimator, an instance of Hutchinson's stochastic trace estimator (Hutchinson, 1990; Avron and Toledo, 2011), whose *number* of automatic-differentiation passes does not grow with the grid.

Proposition 9 (Probe estimators for the squared norms)**.** *Let* $M_* := WA - A^\top W$ *and let* $v \sim \mathcal{N}(0, I)$. *Then*

$$M_* v \;=\; W(Av) \;-\; A^\top (Wv), \qquad \mathbb{E}_v\big[\|M_* v\|^2\big] \;=\; \|M_*\|_F^2, \tag{10}$$

so averaging $\|M_* v\|^2$ *over* m *independent probes is an unbiased estimator of* $\|M_*\|_F^2$, *and the same argument applied to* WA *is unbiased for* $\|WA\|_F^2$. *Each probe costs one Jacobian–vector product* (Av) *and one vector–Jacobian product* $(A^\top u)$, *so the* number *of differentiation passes is* $2m$ *whatever the state dimension.*

Proof. The first identity is linearity: $(WA - A^\top W)v = W(Av) - A^\top(Wv)$, and the two terms are exactly a JVP and a VJP of the shear field, so neither requires A to be formed. For the second, $\mathbb{E}[vv^\top] = I$ gives

$$\mathbb{E}_v\big[\|M_* v\|^2\big] = \mathbb{E}_v\big[v^\top M_*^\top M_* v\big] = \operatorname{tr}\big(M_*^\top M_* \, \mathbb{E}[vv^\top]\big) = \operatorname{tr}\big(M_*^\top M_*\big) = \|M_*\|_F^2.$$

The same argument applied to WA estimates the denominator of (9), so the ratio is formed from two estimates sharing the JVP. □

Remark 10 (What is unbiased, and what the estimator costs). Proposition 9 is unbiased for the two *squared* norms, not for their ratio. What we report is

$$\widehat{\mathcal{D}}_W = \left(\frac{\sum_{i=1}^m \|M_* v_i\|^2}{\sum_{i=1}^m \|WAv_i\|^2} \right)^{1/2}, \tag{11}$$

a ratio of square roots of two unbiased estimates. This is consistent as $m \to \infty$ but is not itself unbiased, and we do not claim otherwise. Numerator and denominator share probes, so their fluctuations are correlated and partly cancel in the ratio; §6.1 measures how much. Two accounting points are worth stating plainly, because it would be easy to overclaim both. First, what is independent of the grid is the *number* of differentiation passes, $2m$ rather than one per degree of freedom; the runtime of each pass still scales with the cost of evaluating the operator, $O(RN^d)$ for the low-rank kernel of §3.4. The saving is a factor of $N^d/2m$ in passes, not a constant-time measurement. Second, the estimator is stochastic, so an $O(1)$ off-diagonal value carries probe noise, which §6.1 quantifies against the dense value; a genuinely symplectic block returns zero for every probe, so the machine-precision diagonal carries none.

3.6 What the theory predicts, and what it leaves open

The results above fix one expectation for each research question of §1, and leave open the matter the experiments exist to settle.

P1, for RQ1. A gradient shear built with the adjoint in W_1 has zero defect in ω_{W_1} at every resolution and dimension (Theorem 5), and non-zero defect in any form not proportional to it (Proposition 7), growing with the spread of $W_2 W_1^{-1}$ and hence with dimension (Remark 8). CKINO's block has no self-adjoint structure and should be non-zero in both forms (Remark 3).

P2, for RQ2. Nothing above says which form helps prediction. Both variants are exactly symplectic; the theory says only that they preserve different forms. The hypothesis RQ2 tests, that the form whose weights are the quadrature of the data's own grid is the one that helps, is a hypothesis about learning, not a consequence of the theorem.

P3, for RQ3. Theorem 5 covers the shears and their compositions. It says nothing about a map that also contains a lift, a projection or a residual update that is not symplectic. Nor does symplecticity imply boundedness under iteration even where it holds: the linear map $(q, p) \mapsto (\lambda q, \lambda^{-1} p)$ with $\lambda > 1$ is symplectic in every ω_W and unbounded. Whether end-to-end symplecticity goes with bounded long rollouts is therefore an empirical question, and RQ3 treats it as one.

4 Architectures

We describe two operators. CKINO is a Chebyshev kernel-integral operator that motivated this study and serves as our non-periodic baseline; SA-Cheb is the self-adjoint construction that makes the symplectic form explicit and controllable.

4.1 CKINO: a Chebyshev kernel-integral operator

CKINO replaces FNO's Fourier multipliers with a low-rank kernel expanded in *Chebyshev coefficients*, so that no periodicity is assumed and the operator can be evaluated on any CGL grid. Figure 3 shows the pipeline; the components are as follows.

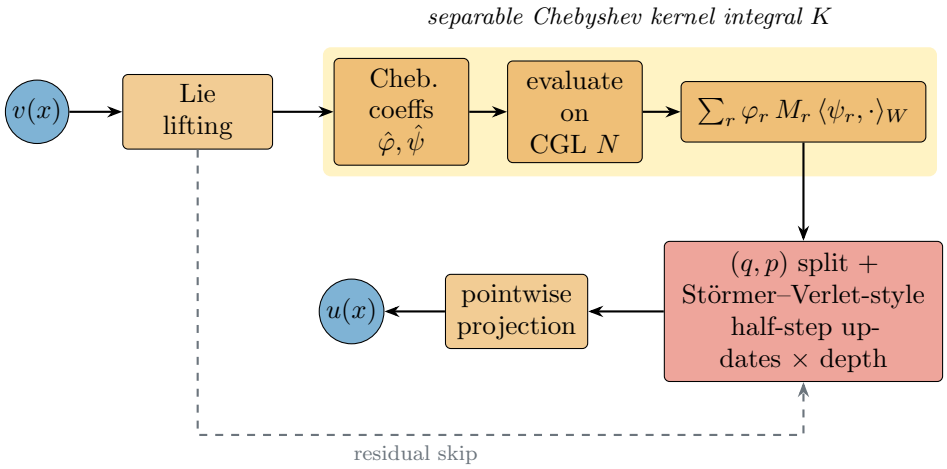


Figure 3: CKINO. The kernel is stored as Chebyshev *coefficients* and evaluated on whatever CGL grid is requested, so the parameters are independent of resolution. The (q, p) block mimics a Störmer–Verlet half-step. Its Jacobian is triangular with unit diagonal, hence volume-preserving; but as §6.1 shows by measurement, *not* symplectic in either candidate form.

(i) Lie-equivariant lifting. A pointwise or short-stencil map raising the C physical fields to H hidden channels, built from generators of a small Lie group so that the lift commutes with translations of the input (Finzi et al., 2020). Two variants matter later: a *convolutional* lift (resolution-dependent stencil) and a *spectral* lift (FFT-derivative, exactly resolution-invariant). §7.5 shows these trade invariance against accuracy.

(ii) Separable Chebyshev kernel integral. The operator core, a low-rank separable kernel in the spirit of Mercer's expansion (Mercer, 1909),

$$(Kv)(x) = \sum_{r=1}^{R} \Big(\prod_{a=1}^{d} \varphi_r^a(x_a) \Big) \sigma_r M_r \langle \prod_a \psi_r^a, v \rangle_W, \tag{12}$$

with φ_r^a, ψ_r^a stored as $n_{\text{train}}+1$ Chebyshev coefficients per axis, $M_r \in \mathbb{R}^{H \times H}$ mixing channels and $\sigma_r > 0$ a learned Mercer scale. Cost is $O(RN^d)$ rather than $O(N^{2d})$, and because the basis functions are polynomials the same weights evaluate on any grid.

(iii) Störmer–Verlet-style block. Hidden channels are split into (q, p) halves and updated by alternating half-steps with a learned step size, in the manner of the Störmer–Verlet scheme (Verlet, 1967; Hairer et al., 2006). The Jacobian of each half-step is unit triangular, so the block is volume-preserving, but it is not symplectic in any of the forms we consider: (12) uses *independent* φ_r and ψ_r and an unconstrained M_r, so nothing forces $k(x, y) = k(y, x)^\top$ and the Jacobian is not self-adjoint (Appendix A).

(iv) Projection. A pointwise map back to C output channels.

4.2 SA-Cheb as an architecture

SA-Cheb keeps the separable low-rank Chebyshev kernel of CKINO (ii) and replaces the Störmer–Verlet-style block (iii) with the gradient shears of §3.4. Each shear field is $F = K^* \rho(Kq)$ (Figure 2),

where K is the low-rank kernel, mapping the c channels of its argument to $h = \max(4, R)$ hidden channels in which the potential acts; K^* is its weighted adjoint in the low-rank form of Appendix B; and ρ is the GELU (Hendrycks and Gimpel, 2016). Shears alternate between updating p from q and q from p, 2ℓ of them at depth ℓ, each scaled by a learned gain initialised at zero so that every shear starts as the identity map. Theorem 5 therefore applies to every block. The lifted variants add a pointwise linear lift and projection (a 1×1 convolution) and a residual update; the lift-free variants add nothing (§4.3). The two forms differ in the weight W, and W enters twice: as the quadrature of the kernel integral inside K, $Kv = \sum_r \varphi_r M_r \langle \psi_r, v \rangle_W$, and, through the same inner product, in K^*. With identical parameters the two forms therefore give different forward maps (a 26% relative difference on a random input at $N = 16$), while $K^* = W^{-1} K^\top W$ holds to 10^{-16} in each. Theorem 5 holds for every K, so it is unaffected; what changes is what the experiments can separate. Changing the form also changes the quadrature rule of the kernel integral, so a model that does better with one form may owe it to either, and we cannot tell which (§9). A control that decouples them, with the kernel quadrature in one weight and the adjoint taken in the other, is straightforward to build; we have not run it.

4.3 Lifted versus lift-free, and why it matters

Theorem 5 is a statement about the *block*. A deployed operator usually wraps blocks in a pointwise lift and projection and is driven residually, $x_{t+1} = x_t + \mathcal{N}(x_t)$. None of those three maps is symplectic in general, so the composition carries no guarantee (Figure 4). We therefore build both:

- **Lifted** (`sacheb`, `sacheb_naive`): $H \gg C$ hidden channels, maximal capacity, symplectic *core* only.
- **Lift-free** (`sacheb_pure`, `sacheb_pure_naive`): the shears act directly on the PDE's own (q, p) fields, $H = C$, no lift, no projection, non-residual update. The *deployed map* is exactly symplectic; capacity must come from rank and depth instead. This requires a canonical (q, p) pair and so applies only to two-field problems.

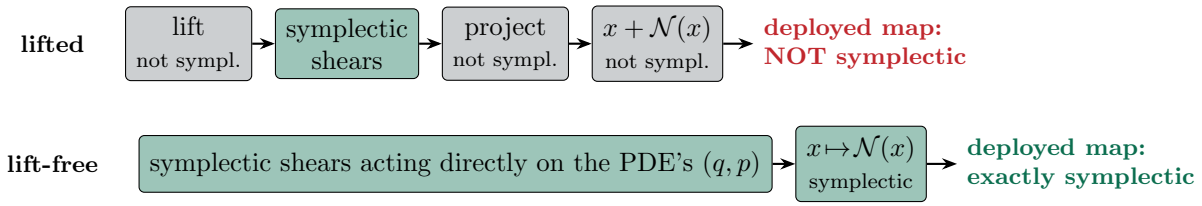


Figure 4: Where the guarantee is lost. The conventional wrapper surrounds a symplectic core with three maps that are not symplectic; the composition therefore carries no guarantee. §6.3 shows this matters in practice: over 2×10^4 steps the lifted variants stay bounded in at most 4 of 12 runs, the matched-form lift-free variant in all 12.

4.4 The variants, and why each exists

Table 2 lists the comparison. The design is a 2×2 factorial, {Chebyshev, Fourier} basis $\times$ {symplectic, not}, realised as SA-Cheb/CKINO/SNO/FNO, crossed with two further controls that vary *which* form is preserved and *whether the deployed map* preserves it.

Table 2: The operator variants compared, and the question each isolates.

variant	symplectic in	role
`sacheb`	W_{cheb} (core only)	Chebyshev-form gradient shear, lifted
`sacheb_naive`	W_{unif} (core only)	identical but uniform-form: isolates *which* form
`sacheb_pure`	W_{cheb} (end to end)	removes the lift: isolates *where* the guarantee holds
`sacheb_pure_naive`	W_{unif} (end to end)	both controls at once; the matched-form, fully-guaranteed operator
`sacheb_kte`, `sacheb_pure_kte`	W_{kte} (core only, end to end)	mapped-grid form (§5.5)
`sacheb_nores_naive`	W_{unif} (core only)	lifted without the residual update: separates the residual from the lift
`sacheb_canon_naive`	W_{unif} (end to end)	the core inside a canonical symplectic wrapper
persistence	any form	the identity map $x \mapsto x$: bounded, no dynamics
CKINO	neither	the non-gradient kernel; volume-preserving only
CKINO-strict	neither	exactly-constrained variant of CKINO
CKINO-nosymp	n/a	structure removed: isolates the block's contribution
SNO (Makara et al., 2026)	W_{unif} (core only)	Fourier symplectic reference
FNO (Li et al., 2021), T-FNO (Kossaifi et al., 2023), U-FNO (Wen et al., 2022), U-Net (Ronneberger et al., 2015), DeepONet (Lu et al., 2021), Transformer (Vaswani et al., 2017)	n/a	unstructured references, our own implementations
GENERIC-FNO (Sulskis and Ravi, 2026)	metriplectic	dissipative reference (see §9)

5 Experimental protocol

5.1 The physical systems

Nine equations span two axes that matter for structure preservation: linear versus nonlinear, and conservative versus dissipative. Four of them are Hamiltonian in a canonical $(q, p) = (u, \partial_t u)$ pair, which is the subset on which a symplectic construction is meaningful at all; the remaining five are scalar fields, where no canonical splitting exists and the lift-free variants of §4.3 cannot be run. Two further Hamiltonian wave problems, sampled on non-uniform nodes, test the matching hypothesis in the other directions (§6.2). Table 3 gives the specification of all eleven systems and Table 4 their initial conditions.

Table 3: The eleven systems: nine in the main matrix and two wave problems on non-uniform nodes. $L = 2$ throughout except `wave1d_dir`, where $L = 1$. "Sub" counts solver micro-steps per stored step. ETD1 and ETD-RK2 are first- and second-order exponential time differencing (Cox and Matthews, 2002); the 2/3 rule removes aliasing from a quadratic nonlinearity (Orszag, 1971). The two-field systems carry a canonical (q, p) pair, and are therefore the only ones on which the lift-free operators of §4.3 can act. † As generated, the nonlinear term of the exponential integrator behind `kdv` carries an extra factor of the micro-step $\Delta t/10 = 10^{-4}$, so the stored trajectories are those of the linear dispersive equation $u_t = u_{xxx}$: their relative L^2 distance to the exact Fourier solution is 2×10^{-4} after 200 stored steps, against 0.43 to the Korteweg–de Vries equation $u_t + 6uu_x - u_{xxx} = 0$ (§9).

Problem	Equation	Domain, boundary	Grid	Δt	Reference integrator
Scalar fields: one channel, no canonical (q, p) pair					
`advection`	$u_t + cu_x = 0$, $c = 1$	$[-1, 1)$, periodic	64	0.005	exact spectral
`heat`	$u_t = \nu u_{xx}$, $\nu = 10^{-2}$	$[-1, 1)$, periodic	64	0.005	exact spectral
`burgers`	$u_t + uu_x = \nu u_{xx}$, $\nu = 2\times10^{-2}$	$[-1, 1)$, periodic	64	0.005	ETD1, 2/3 dealiased, 8 sub
`kdv`†	$u_t = u_{xxx}$ (linear, as generated)	$[-1, 1)$, periodic	64	10^{-3}	exponential, 10 sub
`ns2d`	$w_t + (\mathbf{u}\cdot\nabla)w = \nu\Delta w$, $\nu = 3\times10^{-3}$	$[-1, 1)^2$, periodic	64^2	0.01	ETD-RK2, 2/3 dealiased, 4 sub
Hamiltonian: two channels (u, v), $v = \partial_t u$					
`wave1d`	$u_{tt} = c^2 u_{xx}$, $c = 1$	$[-1, 1)$, periodic	32	0.02	leap-frog, 4 sub
`wave1d_dir`	$u_{tt} = c^2 u_{xx}$, $c = 1$	$[0, 1]$, Dirichlet	64	0.005	leap-frog, 4 sub
`wave2d`	$u_{tt} = c^2 \Delta u$, $c = 1$	$[-1, 1)^2$, periodic	32^2	0.02	leap-frog, 4 sub
`wave3d`	$u_{tt} = c^2 \Delta u$, $c = 1$	$[-1, 1)^3$, periodic	16^3	0.02	leap-frog, 4 sub
`wave1d_cgl`	$u_{tt} = c^2 u_{xx}$, $c = 1$	$[-1, 1]$, Dirichlet	65 CGL	0.01	leap-frog, ≥ 8 sub
`wave1d_kte`	$u_{tt} = c^2 u_{xx}$, $c = 1$	$[-1, 1]$, Dirichlet	65 KTE	0.01	leap-frog, ≥ 8 sub

What each one tests. `advection` is pure transport: linear, non-dispersive, and conserving both $\int u$ and $\int u^2$, so it is the floor below which a surrogate has no excuse. `heat` is linear but strictly dissipative, which makes it a control: energy decay is the correct behaviour, so a model that artificially conserves energy is penalised rather than rewarded. `burgers` adds nonlinearity to dissipation (Burgers, 1948) and steepens into shock fronts, testing whether a spectral parameterisation can represent a sharp front without Gibbs oscillation (Gottlieb and Shu, 1997). `kdv` was intended as the nonlinear dispersive case, the Korteweg–de Vries equation (Korteweg and de Vries, 1895), a Hamiltonian PDE under the non-canonical bracket of Gardner (1971). As generated it is the linear dispersive equation $u_t = u_{xxx}$ (footnote † of Table 3), a conservative problem with the cubic dispersion relation $\omega = k^3$, so that components of different wavelength separate, and with no nonlinear interaction. We keep the identifier `kdv` and report it as the linear dispersive problem; the stiff u_{xxx} term is the reason for $\Delta t = 10^{-3}$ and ten micro-steps. `ns2d` is decaying two-dimensional turbulence in vorticity form, the canonical neural-operator benchmark, and the only problem here with a genuine energy cascade.

The four wave problems carry the conserved Hamiltonian $\mathcal{H} = \frac{1}{2}\int(v^2 + c^2\|\nabla u\|^2)$ and are integrated by Störmer–Verlet leap-frog (Verlet, 1967; Hairer et al., 2006), so the reference trajectories are themselves symplectic. This matters for interpreting §6.3: when we ask whether a learned operator conserves energy over 2×10^4 steps, the ground truth against which it is compared does so by construction.

`wave1d_dir` is the discriminating rung. It is the wave equation on $[0, L]$ with homogeneous *Dirichlet* boundaries, a uniform grid, centred differences and symplectic leap-frog, with initial conditions formed from a truncated Chebyshev series times the envelope $x(L - x)$. The boundary conditions, grid and initial-condition family follow the wave experiment of Makara et al. (2026); the wave speed ($c = 1$ against 0.05), the integrator (leap-frog against RK45), the amount of training data and the model size ($\approx$ 25k parameters against about 2.3M) differ, so our SNO results are not a

reproduction of theirs. The problem is simultaneously Hamiltonian, so symplecticity is meaningful, and non-periodic, so a Fourier parameterisation is structurally mismatched. The reference solver conserves energy to 2.3×10^{-4} over 500 steps and holds the boundary at exactly zero.

Two wave problems on non-uniform nodes. `wave1d_cgl` is the Dirichlet wave equation on $[-1, 1]$ sampled at 65 Chebyshev–Gauss–Lobatto (CGL) nodes, where Clenshaw–Curtis is the correct quadrature and W_{cheb} is the matched form. With the first-derivative matrix D on the nodes and the Clenshaw–Curtis weights W, the semi-discrete operator $A = -W^{-1}D^\top W D$ satisfies $WA = A^\top W$, so $u_t = v$, $v_t = c^2 Au$ is canonically Hamiltonian in ω_W with $H = \frac{1}{2}\big(v^\top W v + c^2 (Du)^\top W (Du)\big)$. The reference trajectories are Störmer–Verlet on the interior nodes with the boundary held at zero; they are symplectic in that form (7×10^{-17} in the relative defect) and H changes by 3×10^{-5} over 500 stored steps. The stored step is 0.01, giving $c\,\Delta t/L = 0.005$ as for `wave1d_dir`. `wave1d_kte` is the same equation on the Kosloff–Tal-Ezer grid (Kosloff and Tal-Ezer, 1993), $x_j = \arcsin(\alpha \xi_j)/\arcsin\alpha$ with $\alpha = 0.995$ and ξ_j the CGL nodes, which relaxes the clustering of the CGL nodes and the time-step restriction that comes with it. Its quadrature is the Clenshaw–Curtis weight times $dx/d\xi$, which is neither constant nor Clenshaw–Curtis, so both standard forms are mismatched there and only a third form W_{kte} is matched.

Table 4: Initial conditions. In every row the coefficients a, b, c, d are drawn independently from $U(-1, 1)$ for each trajectory and each term, and the field is then rescaled so that its maximum absolute value equals the amplitude shown. Initial conditions are drawn from disjoint seed ranges for the training, validation and test splits (§5.3). x, (X, Y) and (X, Y, Z) are the physical coordinates; T_k is the Chebyshev polynomial of degree k.

Problem	Initial state	Amplitude
`advection`	$\sum_{k=1}^{4} a_k \sin(\pi k x) + b_k \cos(\pi k x)$	0.5
`heat`	$\sum_{k=1}^{6} a_k \sin(\pi k x)/k$	0.6
`burgers`	$\sum_{k=1}^{3} a_k \sin(\pi k x) + b_k \cos(\pi k x)$	0.6
`kdv`	$\sum_{k=1}^{4} c_k \sin(\pi k x)$, rescaled to amplitude 0.3, plus 0.5	0.3
`ns2d`	vorticity $\sum_{k_x,k_y=1}^{4} \big[a \sin(\pi k_x X)\cos(\pi k_y Y) + b \cos(\pi k_x X)\sin(\pi k_y Y)\big]$, mean removed	0.8
`wave1d`	$u = \sum_{k=1}^{3} c_k \cos(\pi k x)$, $v = \sum_{k=1}^{3} d_k \sin(\pi k x)$	0.4 each
`wave2d`	$u = \sum_{k_x,k_y=1}^{2} a \cos(\pi k_x X)\cos(\pi k_y Y)$, $v = \sum_{k_x,k_y=1}^{2} b \sin(\pi k_x X)\sin(\pi k_y Y)$	0.4 each
`wave3d`	$u = \sum_{k_x,k_y \in \{1,2\}} a \cos(\pi k_x X)\cos(\pi k_y Y)\cos(\pi Z)$, $v = \sum_{k_x,k_y \in \{1,2\}} b \sin(\pi k_x X)\sin(\pi k_y Y)\sin(\pi Z)$	0.4 each
`wave1d_dir`	$x(L-x)\sum_{k=0}^{3}(1+k)^{-2} c_k T_k(2x/L-1)$ for u and for v (independent coefficients), zero at both ends	0.4, 0.05
`wave1d_cgl`, `wave1d_kte`	$(1-x^2)\sum_{k=1}^{4} a_k T_k(x)$ for u and v (independent coefficients), evaluated at the CGL or mapped nodes, zero at both ends	0.4 each

5.2 Nodes, representation and quadrature in each experiment

The theory of §3 is posed on CGL nodes, where the Clenshaw–Curtis weights are the correct quadrature. The experiments are not: nine of the eleven datasets of Table 3 are sampled on a uniform grid, and the other two on CGL and mapped nodes. Table 5 states exactly what this

means for each run, because the paper's central claim is that the inner product must match the discretisation, and a reader cannot check that claim without knowing which nodes carry which weights.

Table 5: Nodes and quadrature in every experiment. *Physical nodes*: where the data were sampled, a uniform grid periodic on $[-1, 1)^d$ except for `wave1d_dir`, which includes both ends of $[0, 1]$, and the last two rows, which are Dirichlet problems on non-uniform nodes. *Model-evaluation nodes*: where the operator evaluates its Chebyshev basis; the j-th of n samples is read as the CGL node $x_j = \cos(\pi j/N)$, $N = n - 1$, with no resampling. *Reference quadrature*: the weights of the solver's own discrete L^2 product (trapezoid rule, spacing h). CC_n is the diagonal of Clenshaw–Curtis weights for n CGL nodes, $\approx \frac{\pi}{N} \sin \frac{\pi j}{N}$ in the interior and $O(N^{-2})$ at the two ends; in d dimensions every weight is the tensor product of its per-axis factor. The reference quadrature is proportional to W_{unif} in the first nine rows, to W_{cheb} in `wave1d_cgl`, and to $W_{\mathrm{kte}} = \mathrm{CC}_{65}\, dx/d\xi$ in `wave1d_kte`, where ξ_j are the CGL nodes and x_j the mapped nodes.

Dataset	Physical nodes	Model-evaluation nodes	Reference quadrature	W_{cheb} definition	W_{unif} definition
`advection`	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
`heat`	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
`burgers`	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
`kdv`	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
`wave1d`	32 uniform, periodic	32 CGL, $N = 31$	$h = 2/32$	CC_{32}	$\frac{2}{32} I$
`wave1d_dir`	64 uniform, ends incl.	64 CGL, $N = 63$	$h = 1/63$	CC_{64}	$\frac{2}{64} I$
`wave2d`	32^2 uniform, periodic	32^2 CGL	h^2, $h = 2/32$	$\mathrm{CC}_{32}^{\otimes 2}$	$(\frac{2}{32})^2 I$
`wave3d`	16^3 uniform, periodic	16^3 CGL	h^3, $h = 2/16$	$\mathrm{CC}_{16}^{\otimes 3}$	$(\frac{2}{16})^3 I$
`ns2d`	64^2 uniform, periodic	64^2 CGL	h^2, $h = 2/64$	$\mathrm{CC}_{64}^{\otimes 2}$	$(\frac{2}{64})^2 I$
`wave1d_cgl`	65 CGL, Dirichlet	65 CGL, $N = 64$	CC_{65}	CC_{65}	$\frac{2}{65} I$
`wave1d_kte`	65 KTE, Dirichlet	65 CGL, $N = 64$	$\mathrm{CC}_{65}\, dx/d\xi$	CC_{65}	$\frac{2}{65} I$

The table answers the questions a reader needs settled before the ablation can be read.

Is anything resampled? No. Each uniform sample is handed to the model as the value at the corresponding CGL node, without interpolation, and both variants receive identical inputs.

What does the operator then represent? Because $T_k(\cos\theta) = \cos k\theta$, the Chebyshev basis evaluated at node j is $T_k(x_j) = \cos(k\pi j/N)$. In the physical coordinate $s_j = j/N$ of the uniform grid this is $\cos(k\pi s_j)$, so on these data the operator's kernel functions are cosine series on a uniform grid. The parameterisation never assumes periodicity, which is what matters for the boundary results of §7.1, but the operator is not acting on data sampled at Chebyshev nodes.

Which inner product is the physical one? The solver's quadrature, $\langle f, g\rangle \approx h \sum_j f_j g_j$, which is proportional to W_{unif}: exactly so on the periodic grids, and on every non-zero entry for `wave1d_dir`, whose boundary values vanish. A constant rescaling of W does not change which maps preserve the form (Proposition 7 with $W_2 = cW_1$), so W_{unif} is the matched form.

What, then, does W_{cheb} do? It attaches the Clenshaw–Curtis weights, approximately $\frac{\pi}{N}\sin(\pi s_j)$, to uniformly spaced samples. That is a legitimate inner product, the quadrature of $\int fg\, \pi \sin(\pi s)\, ds$, but not the physical one: it weights the middle of the domain up to $\approx \pi N$ times more than the ends.

Where do the two forms enter? In the quadrature $\langle \psi_r, v\rangle_W$ of the kernel integral and, through the same inner product, in the adjoint $K^* = W^{-1}K^\top W$ of (3) (§4.2). The basis, rank, channel mixing, initialisation and inputs are identical.

The consequence is worth stating, since it bounds what the nine-problem experiments can establish. Because every one of the nine matrix grids is uniform, W_{unif} is the matched form throughout and W_{cheb} the mismatched one throughout, so they test the matching hypothesis in one direction only: they cannot by themselves separate "matching the form to the grid is what matters" from "constant weights are simply the better choice". The two further Hamiltonian problems of

Table 3, one sampled on CGL nodes where W_{cheb} is the matched form and one on a mapped grid where neither standard form is, test the other directions (§6.2).

5.3 Data, splits and selection

Trajectories are generated from random initial conditions with *disjoint seed ranges* per split, so no initial condition appears in more than one split. Each 1-D problem uses 512 training trajectories of 600 steps and is trained for 15 epochs; `wave2d` and `ns2d` use 96 trajectories of 300 steps (15 epochs), and `wave3d` uses 48 trajectories of 200 steps (12 epochs). In the main matrix every problem has 12 validation and 12 test trajectories; the additional runs of §5.5 use 32 test trajectories. The data seeds are fixed independently of the training seed, so every model and every seed is scored on the *same* test trajectories; the training seeds vary only initialisation and minibatch order. Rollout metrics are always computed on the test split, and are evaluated once.

Training. All models are trained with AdamW (Loshchilov and Hutter, 2019), a variant of Adam (Kingma and Ba, 2015) with decoupled weight decay (10^{-5}), at a peak learning rate of 3×10^{-3} annealed by a cosine schedule (Loshchilov and Hutter, 2017) to 3×10^{-5}, with gradient-norm clipping at 1 and batches of 32. Recursive models are trained by push-forward unrolling over horizons of 1, 2 and 4 steps in equal thirds of the epochs, with teacher forcing decaying from 1 to 0 and, in the noisy configurations, input noise of relative size 2%; non-recursive `seq2seq` models predict a window of future states from the initial state in one pass, with the number of gradient steps matched to the recursive models. Everything is implemented in PyTorch (Paszke et al., 2019); the baselines are our own implementations from the published descriptions. The long-horizon and multi-resolution runs use the same optimiser, schedule and unrolling curriculum, without input noise, and 12 epochs.

One caveat applies to Table 14 specifically. The best configuration per problem is the one with the lowest mean over three seeds of test-split relative RMS at rollout step 200 (150 in 3-D), compared across all configurations; validation is not used for this selection. Selecting the minimum of many noisy scores biases it optimistically (Cawley and Talbot, 2010), so the per-problem winners should be read as an upper estimate rather than an unbiased one. The comparisons the three research questions rest on are not exposed to this: the defect matrix (§6.1), the form ablation (§6.2) and the long-horizon rollouts (§6.3) are all paired comparisons between variants that differ in one design decision (the weight W, or the presence of the lift), with no selection over a pool.

5.4 Design, matching and cost

Families are crossed with prediction mode {recursive, non-recursive `seq2seq`}, training noise {none, 2%} and an optional PDE-residual loss, giving 33 configurations in 1-D and 21 in 2-D/3-D, all matched to ≈25k parameters. Three seeds on a single accelerator type (NVIDIA Tesla T4): **783 training runs**. The additional runs of §5.5 add 338.

Parameter count is the matching criterion, and it is the weakest of the available ones. The families compared here have very different arithmetic profiles at equal parameter count: a Fourier layer spends its budget on multipliers applied through an FFT of cost $O(N^d \log N)$, a low-rank integral kernel on $O(RN^d)$ contractions, attention on a quadratic term in sequence length, and a U-Net on convolutions across a resolution hierarchy. Autoregressive rollout multiplies all of these by the number of steps, while the `seq2seq` variants pay once. Equal parameters therefore does not mean equal memory, equal FLOPs or equal wall-clock, and none of the accuracy comparisons in §7.2 should be read as statements about efficiency. Measured throughput is reported separately in §7.5, where the spread is roughly an order of magnitude.

Metrics. Relative RMS at rollout checkpoints, plus amplitude ratio, pattern correlation, spectral error and invariant drift, collapsed into a *verdict* and a *usable horizon* (last step with correlation ≥ 0.9 and amplitude ratio in $[0.7, 1.4]$). A model can post a middling RMS while being physically dead, and only the decomposition separates the two. For the long-horizon rollouts of §6.3 we report four of these separately, each averaged over the 8 test trajectories of those runs (§5.5): relative RMS against the reference, the measure of *trajectory* error; pattern correlation, the Pearson correlation of the mean-removed predicted and reference fields, which measures *phase*; amplitude ratio, the ratio of their standard deviations; and energy drift $|E(t) - E(0)|/E(0)$, computed with the solver's own energy functional on the model's own trajectory.

5.5 Additional seeds, grids and controls

Further experiments extend the main matrix in four ways and address the limitations stated in §5.2 and §9.

Two further grids. The wave problems `wave1d_cgl` and `wave1d_kte` of §5.1 test the matching hypothesis in the other directions: on CGL nodes W_{cheb} is the matched form, and on the mapped grid neither standard form is. As before, the operator reads sample j as the CGL node ξ_j.

Seeds, batches and uncertainty. The lifted and lift-free pairs of the nine uniform problems were run with five seeds (0–4); `wave1d_cgl` and `wave1d_kte` used five throughout. Seeds 0–2 belong to the main matrix, which ran on PyTorch 2.8 and scored 12 test trajectories per run; seeds 3–4 and the two further grids ran on PyTorch 2.13 and scored 32 test trajectories per run, from the same split seeds. The two batches therefore differ in software and in test-set size, and we pool them only through the per-seed error. For each pair we take, per seed, the ratio of the mismatched to the matched model's relative RMS at step 200 (150 in 3-D), with errors above 10^3 clipped as in §5.3, and report the number of seeds won and the median ratio with its range. A comparison is called only if a 95% percentile bootstrap interval (Efron and Tibshirani, 1994) for the mean log-ratio over seeds excludes zero *and* the median ratio differs from one by at least 5%; otherwise it is reported as inconclusive. The 5% floor is an analysis choice made after inspecting the data, not a pre-specified one: it was introduced when a 1% difference was flagged by an interval whose lower end was barely above one while three of five seeds agreed. It can only turn a call into an inconclusive one, and it changes none of the uniform-grid or CGL calls. The intervals are descriptive: with five seeds they are not a substitute for a formal test.

Long-horizon protocol and controls. Each long-horizon model is trained on 384 trajectories of 400 steps for 12 epochs with a recursive curriculum, then rolled out for 2×10^4 steps from the first 8 test initial conditions, scoring at log-spaced checkpoints; every number is the mean over those 8 trajectories. The families of Table 10 were run with three seeds each. A second batch repeats the lift-free and lifted W_{unif} families on the four wave problems and SNO and FNO on the two 1-D problems, with the same seeds; runs the lifted and lift-free families on the two further grids; and adds controls. *Canonical wrapper*: the core inside a learned pointwise change of coordinates ($q \mathrel{+}= Ap$, $p \mathrel{+}= Bq$, $q \mathrel{+}= Cp$ with symmetric A, B, C, and its inverse), which acts on channels only and is therefore symplectic in every diagonal W; it adds three parameters. *No residual*: the lifted variant without the residual update. *Persistence*: $x \mapsto x$. *Untrained*: the lift-free core with its gains fixed at 0.3. *One epoch*: the same core trained for a single epoch. *Perturbed adjoint*: the adjoint's channel-mixing matrix is replaced by $M + \epsilon\|M\|\,\Delta/\|\Delta\|$ for a fixed random Δ, which is

symplectic in no form. At the first test state we also measure the end-to-end defect (float64 in 1-D, float32 otherwise) and the eigenvalues of the one-step Jacobian (1-D and 2-D).

Unseen resolutions. On `wave1d_dir` and `wave1d_cgl` we train on two resolutions (32 and 64 points, or 33 and 65 nodes) with 256 trajectories each, and test the one-step error on six, four of them unseen (24, 48, 96, 128, or 25, 49, 97, 129), with 3 seeds.

6 Results: the three research questions

6.1 RQ1: Which form does each construction preserve?

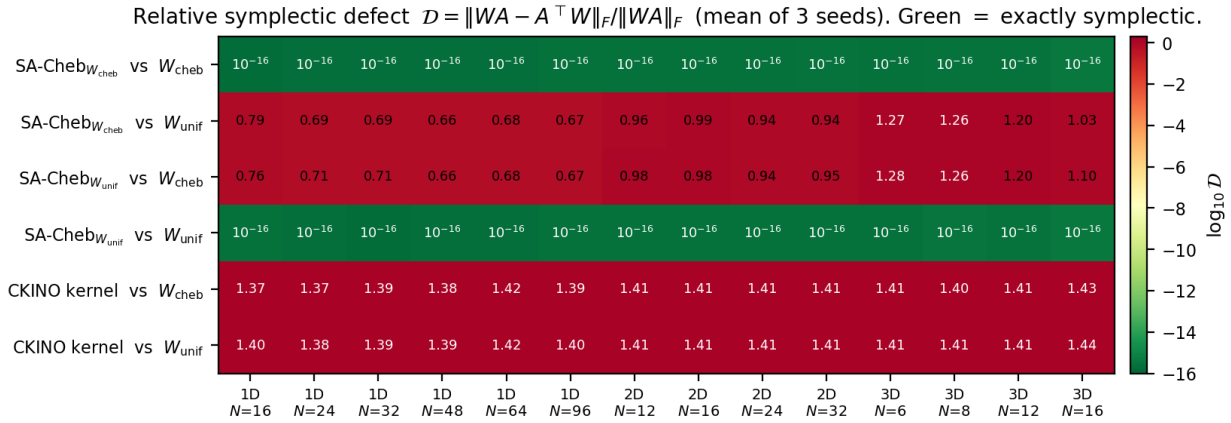


Figure 5: Symplectic defect (9) of every construction against *both* inner products, over 14 grid configurations in 1-D, 2-D and 3-D, mean of three independently initialised operators. Every cell uses the dense Jacobian except the largest 3-D grid (17^3 nodes), which uses the probe estimator with $m = 96$. The green diagonal is the result: each gradient shear is exact in the form its adjoint was built from and $O(1)$ in the other. The CKINO kernel is ≈ 1.4 in both: volume-preserving, not symplectic.

Figure 5 is the central measurement, and it bears out P1 on every count.

(i) Theorem 5 holds numerically: the matched-form defect is at most 4.4×10^{-16}, independent of resolution and dimension. **(ii)** The two variants are *both* exactly symplectic, in different forms; neither is "broken". **(iii)** The off-diagonal grows with dimension, as Remark 8 predicts: averaged over grids, models and seeds it is 0.70 in 1-D (range 0.44–1.02), 0.96 in 2-D (0.80–1.11) and 1.20 in 3-D (0.97–1.35). **(iv)** The CKINO kernel is 1.3–1.5 in both forms, the conflation flagged in Remark 3.

Dense against probe. Every grid with at most 4096 degrees of freedom, which is all but one, is computed from the dense Jacobian; only the $17^3 = 4913$-node grid uses the estimator (11). To see how much that estimate can be trusted we compared it with the dense value on grids where both are affordable (65, 25^2 and 13^3 nodes), drawing 40 independent probe sets per cell (Table 6). At the $m = 96$ we use, the mean estimate is within 1% of the dense value, which is inside the Monte Carlo error of 40 repeats, so there is no detectable bias. The probe-to-probe standard deviation is 3.8–4.4%, and 95% of single estimates fall within 9% of the dense value. The spread falls roughly as $m^{-1/2}$, and sharing probes between numerator and denominator, as (11) does, makes it 1.3–2× smaller than drawing them independently. The one probe-estimated cell of Figure 5 therefore carries

about ±4% (one standard deviation) of probe noise on a value near 1.2, which affects no comparison in this paper. On the diagonal every probe returns machine zero, so there is no noise to report.

Table 6: Probe estimator against the dense defect: 40 independent probe sets per cell, off-diagonal entries of both SA-CHEB variants, worst of the two shown. *SD*: probe-to-probe standard deviation relative to the dense value. *p95*: 95th percentile of $|\widehat{\mathcal{D}}/\mathcal{D} - 1|$. Shared probes use the same v_i in numerator and denominator, as in (11).

		1-D, 65 nodes		2-D, 25^2		3-D, 13^3	
m	probes	SD	p95	SD	p95	SD	p95
8	shared	13.7%	25.7%	22.2%	46.8%	16.9%	29.3%
32	shared	6.7%	13.8%	10.3%	16.4%	7.6%	14.7%
96	shared	3.8%	7.7%	4.4%	9.0%	4.3%	8.4%
96	independent	4.9%	8.4%	9.0%	15.8%	7.9%	15.9%

6.2 RQ2: Does matching the form to the grid improve prediction?

Every pair compared here is identical in architecture, parameter count and training recipe, and differs only in the weight W, which enters both the kernel's quadrature and its adjoint (§4.2). The matched form is the one whose weights are the quadrature of the data's own grid: W_{unif} on the nine uniform grids, W_{cheb} on CGL nodes, and W_{kte} on the mapped grid, where neither standard form is. Figure 6 shows the per-seed evidence for all three; the protocol is in §5.5.

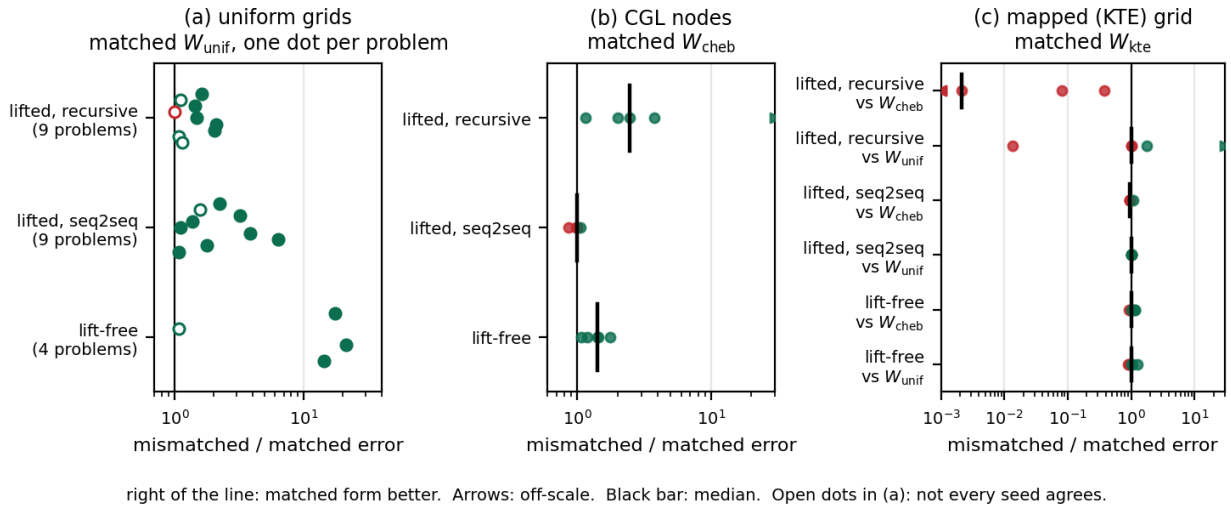


Figure 6: Matched versus mismatched form at identical capacity: error ratio mismatched/matched at step 200 (150 in 3-D), five seeds, errors clipped at 10^3. A ratio above one means the matched form is better. **(a)** Uniform grids, matched W_{unif}: one dot per problem (median over seeds); open dots mark problems on which not every seed agrees. **(b)** CGL nodes, matched W_{cheb}: one dot per seed. **(c)** Mapped grid, matched W_{kte}: one dot per seed, against each of the other two forms. Black bars are medians; arrows mark seeds off the plotted range.

Uniform grids. Table 7 summarises the 22 comparisons that the lifted and lift-free configurations give on the nine problems (the lift-free pair exists only on the four two-field problems). The matched

form is called better in 17, inconclusive in 5, and worse in none. Sixteen comparisons favour it in every one of the five seeds and five in four of five. The five inconclusive ones are both lifted pairs on heat, and the recursive lifted pair on `wave3d`, `ns2d` and `wave1d_dir`; in the first four the matched form still leads on the median (1.08–1.56×), and in `wave1d_dir` nearly every run of both models diverges, so the ratio is 1.00. Seeds 3 and 4, run on PyTorch 2.13 with a larger test set, agree with seeds 0–2: the matched form wins 40 of 44 seed pairs.

Table 7: Uniform grids, matched W_{unif} against mismatched W_{cheb}, five seeds. *Matched*, *inconclusive* and *other* count problems by the verdict rule of §5.5; *seeds won* counts seed pairs in which the matched form has the lower error; the last column is the median over problems of the per-problem median ratio, with the range of those medians.

comparison	problems	matched	inconclusive	other	seeds won	median ratio [range]
lifted, recursive	9	5	4	0	39/45	1.44 [1.00, 2.09]
lifted, seq2seq	9	8	1	0	44/45	1.77 [1.07, 6.30]
lift-free	4	4	0	0	19/20	15.87 [1.07, 21.26]
all uniform	22	17	5	0	102/110	

For the lift-free pair (Table 8) the matched form is better by 17, 21 and 14× on `wave1d`, `wave2d` and `wave3d` in every seed, and by 1.07× on `wave1d_dir` (four seeds of five). On the first three the mismatched model has failed outright, with an error near the signal (0.9–1.5) against 0.04–0.09, so those ratios contrast a working model with a failed one rather than grading two working models.

Table 8: Lift-free pair, both models exactly symplectic *end to end*, differing only in the form. Relative RMS at step 200 (150 in 3-D), median over five seeds.

problem	seeds	`purecheb` (W_{cheb})	`pureunif` (W_{unif})	ratio	[range]	seeds won
`wave1d`	5	1.5219	0.0880	17.34	[14.97, 20.06]	5/5
`wave1d_dir`	5	0.1050	0.0877	1.07	[0.98, 1.44]	4/5
`wave2d`	5	0.9047	0.0414	21.26	[16.09, 22.78]	5/5
`wave3d`	5	1.0568	0.0734	14.40	[10.60, 16.91]	5/5

The reverse direction. On CGL nodes the roles of the two standard forms swap, and the matching hypothesis predicts that the preference reverses. For two of the three pairs it does (Table 9). With the lifted recursive configuration W_{cheb} wins in all five seeds, by a median of 2.45× (the largest ratio, 53×, comes from a seed in which the W_{unif} model diverged), and with the lift-free pair it wins in all five, by a median of 1.42×; the trajectory-level geometric mean is 1.35× with a 95% bootstrap interval of $[1.16, 1.56]$. For the non-recursive (`seq2seq`) pair there is no preference: 0.99×, with W_{cheb} ahead in one seed of five. The reversal is smaller than the forward effect: 1.4× for the lift-free pair against 14–21× on the uniform grids. One possible reason, which we have not tested, is that the mismatched model fails more gently here: the CGL fields vanish at the clustered end nodes where W_{unif} and W_{cheb} differ most, whereas on the periodic uniform grids the signal has support everywhere.

A grid where neither standard form is right. On the mapped grid the exactly matched W_{kte} brings no consistent benefit. The `seq2seq` and lift-free pairs are within 4% of each other for all three weights and inconclusive. For the lifted recursive pair W_{kte} is worse than W_{cheb} in all five

Table 9: Reverse direction and mapped grid, five seeds. *vs* names the mismatched form the matched one is compared with; *seeds won* counts seeds in which the matched form has the lower error; ratios are mismatched/matched (median and range over seeds); *verdict* follows the rule of §5.5, where "other form" means the mismatched form is better.

grid (matched)	comparison	vs	seeds won	ratio [range]	verdict
CGL (W_{cheb})	lifted, recursive	W_{unif}	5/5	2.45 [1.15, 52.82]	matched form
	lifted, seq2seq	W_{unif}	1/5	0.99 [0.85, 1.05]	inconclusive
	lift-free	W_{unif}	5/5	1.42 [1.06, 1.75]	matched form
mapped (W_{kte})	lifted, recursive	W_{cheb}	0/5	2.1×10^{-3} [2.6×10^{-4}, 0.38]	other form
	lifted, recursive	W_{unif}	2/5	1.00 [0.01, 80.51]	inconclusive
	lifted, seq2seq	W_{cheb}	1/5	0.96 [0.95, 1.07]	inconclusive
	lifted, seq2seq	W_{unif}	4/5	1.01 [1.00, 1.02]	inconclusive
	lift-free	W_{cheb}	3/5	1.01 [0.94, 1.16]	inconclusive
	lift-free	W_{unif}	3/5	1.01 [0.89, 1.25]	inconclusive

seeds and indistinguishable from W_{unif}, but that comparison is dominated by failures: at step 200 the error exceeds 3 in four of five seeds for W_{kte}, in all five for W_{unif}, and in one for W_{cheb}. The mapped grid therefore does not support "the matched form is best" in any of the pairs.

What this establishes. The preference follows the grid where the grid has a standard quadrature: toward W_{unif} on uniform grids (17 of 22 comparisons, none against) and toward W_{cheb} on CGL nodes (two of three, none against). That is the reversal the matching hypothesis needed to survive, and it did. It is not universal, since the non-recursive CGL pair is a tie and the mapped grid shows nothing, and it is limited by a confound. The weight also defines the quadrature of the kernel integral, so what was varied is the weighting of the *whole operator*, not the symplectic form alone. The data therefore support "use the grid's quadrature weights" at least as directly as "preserve the grid's symplectic form", and the control that would separate them (kernel quadrature fixed, adjoint weight varied) is the main experiment not performed here (§9). The answer to RQ2 is therefore: yes where the grid has a standard quadrature, in both directions, with effects from 1 to 20×; no on the mapped grid; and not yet attributable to the symplectic form specifically.

6.3 RQ3: End-to-end symplecticity and the conventional wrapper

Each long-horizon run trains a model once and rolls it out autoregressively for 2×10^4 steps from the first 8 test initial conditions, on the four wave problems with three seeds each. With $c = 1$ and $\Delta t = 0.02$ on $[-1, 1)^d$ that is 400 time units, or 200 transits of the domain; for `wave1d_dir` ($\Delta t = 0.005$, $L = 1$) it is 100 transits. Nothing at this horizon is visible in the few-hundred-step main matrix. Following §5.4, we keep apart four things that a single error number would conflate: whether the state stays *bounded*; how far it is from the reference *trajectory*; whether it keeps the right *phase* and *amplitude*; and whether it conserves *energy*. We call a run bounded if its relative RMS at step 2×10^4 is at most 3. The threshold does not matter: the largest error among bounded runs is 1.99 and the smallest among the rest is 4.68, so any threshold in between gives the same classification.

Boundedness. The matched-form lift-free operator is the only family of the main study bounded in every run (Table 10). The same construction in the mismatched form is bounded on the three lower-dimensional problems and diverges on `wave3d` in every seed, although it is exactly symplectic

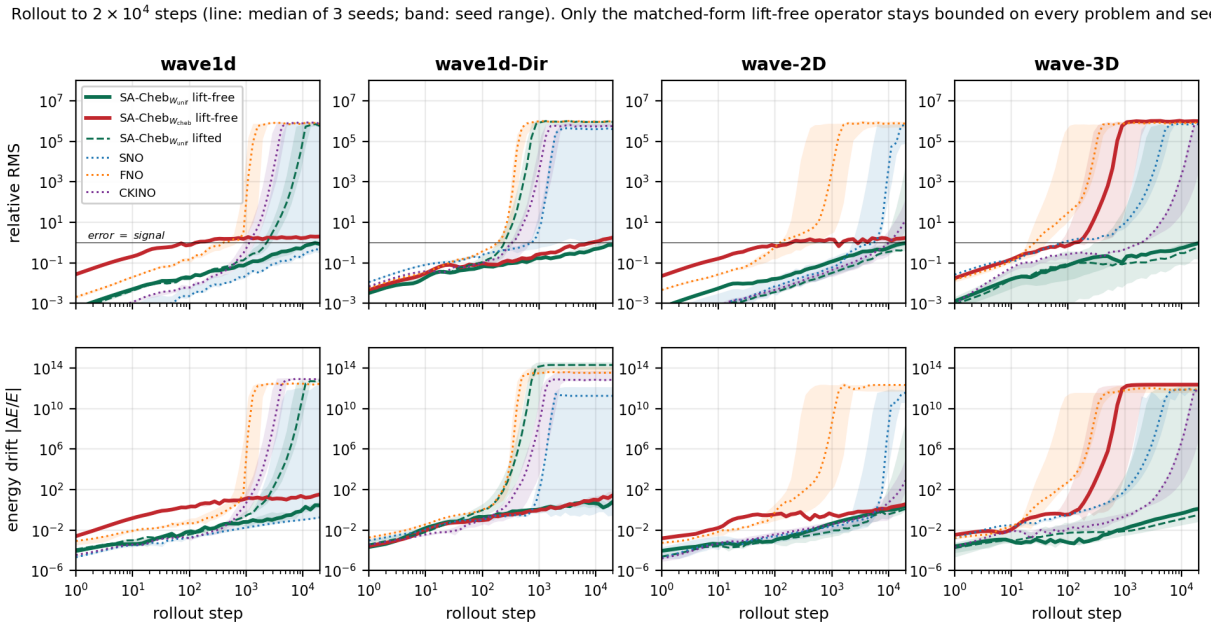


Figure 7: Rollout to 2×10^4 steps on the four wave problems; line: median of 3 seeds, band: range over seeds. **Top:** relative RMS against the reference; the thin line marks error = signal, and diverged runs saturate near 10^6, where the rollout clamps the state. **Bottom:** energy drift $|\Delta E/E|$ on the model's own trajectory. The matched-form lift-free operator (solid green) is the only one whose band stays bounded on every problem. The mismatched-form lift-free operator (solid red) diverges on `wave3d` in every seed. The lifted W_{unif} variant (dashed green) has a low median on `wave2d` and `wave3d` because two of its three seeds stay bounded there; its band shows the third diverging.

Table 10: Long-horizon census: runs still bounded (relative RMS ≤ 3) at step 2×10^4, out of three seeds per problem. *Symplectic* says where the guarantee of Theorem 5 holds. T-FNO was run on the 1-D problems only.

family	symplectic	`wave1d`	`wave1d_dir`	`wave2d`	`wave3d`	total
`pureunif` (lift-free, W_{unif})	end to end, matched	3	3	3	3	**12/12**
`purecheb` (lift-free, W_{cheb})	end to end, mismatched	3	3	3	0	9/12
SA-Cheb lifted, W_{unif}	core only	0	0	2	2	4/12
SA-Cheb lifted, W_{cheb}	core only	0	0	2	1	3/12
SNO	core only	2	1	0	0	3/12
CKINO-strict	no	0	0	1	0	1/12
CKINO	no	0	0	0	0	0/12
T-FNO	no	1	0	n/a	n/a	1/6
FNO	no	0	0	0	0	0/12

end to end: as P3 anticipated, exact symplecticity is not sufficient for boundedness. The lifted SA-Cheb variants, whose symplectic shears sit inside a non-symplectic lift, projection and residual update, are bounded in 4 and 3 runs of 12, and never on a 1-D problem; SNO, also a symplectic core inside a lift, in 3; the CKINO variants in at most 1; FNO in none.

Accuracy of the runs that stay bounded. Boundedness is not accuracy, and the two orderings differ (Table 11). The matched-form lift-free runs end with relative RMS 0.77–0.95 (median over seeds), an error comparable to the signal itself, and pattern correlation 0.41–0.65: they keep the amplitude of the field (ratio 1.00–1.19) but not its phase. When a lifted or Fourier operator does stay bounded it is *more* accurate: the lifted W_{unif} variant reaches relative RMS 0.44 on `wave2d` and 0.35 on `wave3d`, with median pattern correlation above 0.9, and SNO reaches 0.36 on `wave1d` with correlation 0.94. What end-to-end symplecticity bought in these experiments is reliability, in that the matched-form model never diverged, rather than accuracy; the controls below show how little that reliability alone establishes.

Table 11: State at step 2×10^4 of the runs that stayed bounded: median over those runs, whose number is given in the third column; families with no bounded run on a problem are omitted (Table 10). Relative RMS measures trajectory error, pattern correlation phase, the amplitude ratio size, and $|\Delta E/E|$ energy conservation on the model's own trajectory. The reference solver's energy drift over the same horizon is at most 3.3×10^{-4}.

problem	family	bounded	rel. RMS	pattern corr.	amp. ratio	$\lvert\Delta E/E\rvert$
`wave1d`	`pureunif`	3/3	0.84	0.65	1.04	2.73
	`purecheb`	3/3	1.98	0.05	1.54	33.6
	SNO	2/3	0.36	0.94	1.03	0.16
	T-FNO	1/3	1.45	0.16	1.31	1.11
`wave1d_dir`	`pureunif`	3/3	0.77	0.41	1.00	6.25
	`purecheb`	3/3	1.77	−0.34	1.28	27.7
	SNO	1/3	1.31	−0.69	0.79	0.29
`wave2d`	`pureunif`	3/3	0.95	0.59	1.09	2.48
	`purecheb`	3/3	1.70	−0.32	1.17	3.64
	lifted, W_{unif}	2/3	0.44	0.91	1.09	1.07
	lifted, W_{cheb}	2/3	0.98	0.78	1.48	5.49
	CKINO-strict	1/3	1.92	0.54	2.26	38.7
`wave3d`	`pureunif`	3/3	0.94	0.65	1.19	1.26
	lifted, W_{unif}	2/3	0.35	0.93	1.09	0.20
	lifted, W_{cheb}	1/3	1.21	0.51	1.35	0.85

Which part of the wrapper loses boundedness? Variants between the bare core and the conventional wrapper separate the stages (Table 12, Figure 8). The core alone is bounded in all 18 runs, including the two non-uniform grids. Inside a learned canonical wrapper it stays bounded (12 of 12). That the wrapped map is symplectic is guaranteed, since a composition of symplectic maps is symplectic; that it stays bounded is an experimental result, not a consequence. Behind a pointwise linear lift and projection, with no residual update, it is bounded in 3 of 12 runs; with the residual update added, the conventional wrapper, in 4 of 18 (0 of 6 on the two non-uniform grids). The exception is `wave2d`, where the full wrapper stays bounded in all three seeds and is the most accurate variant (median error 0.58 against 0.66 for the core). So the loss of boundedness comes

with the non-symplectic lift and projection, and the residual update adds little in 1-D and 3-D. This is not a controlled separation of symplecticity from capacity: the canonical wrapper adds three parameters, while the lifted variants have a different width and rank at the same 25k budget, so "symplectic wrapper versus not" is crossed with the size of the wrapper.

Table 12: Long-horizon census of the staged ablation and the controls: runs bounded (relative RMS ≤ 3 at step 2×10^4) out of three seeds per problem. The first block is the matched-form lift-free operator and its lifted counterpart (the same seeds as Table 10, run again in the second batch); later blocks are controls. *Symplectic* says what the construction guarantees. A dash marks a variant not run on that problem; SNO and FNO were run on the two 1-D problems.

variant	symplectic	`wave1d`	`wave1d_dir`	`wave2d`	`wave3d`	`cgl`	`kte`	total
core only (lift-free), W_{unif}	end to end	3	3	3	3	3	3	18/18
core + canonical wrapper	end to end	3	3	3	3	–	–	12/12
core + linear lift/projection	core only	1	0	1	1	–	–	3/12
core + lift/projection + residual	core only	0	0	3	1	0	0	4/18
persistence ($x \mapsto x$)	trivially	3	3	3	3	3	3	18/18
core, trained one epoch	end to end	3	3	–	–	–	–	6/6
core, untrained	end to end	0	0	–	–	–	–	0/6
core, adjoint perturbed 1%	approximately	3	3	–	–	–	–	6/6
core, adjoint perturbed 10%	approximately	3	3	–	–	–	–	6/6
core, adjoint perturbed 100%	no	0	0	–	–	–	–	0/6
SNO	core only	2	2	–	–	–	–	4/6
FNO	no	0	0	–	–	–	–	0/6

Everything that is exactly symplectic end to end. Across both batches, 63 rollouts used a map that is exactly symplectic end to end in some form: the lift-free operators of Table 10 in both forms (24), the core in the three forms of the second batch (27, of which 12 re-run the matched-form operator of the first batch on the same seeds) and the canonical wrapper (12). Sixty were bounded. The three that were not are the mismatched-form model on `wave3d`, in every seed; the other mismatched-form runs, on `wave1d`, `wave1d_dir`, `wave2d`, the CGL grid and the mapped grid, were bounded.

What boundedness does and does not show. Three controls bound the reading. First, the identity map $x \mapsto x$ is exactly symplectic, has zero energy drift, and is bounded in all 18 runs, so boundedness alone is no evidence of dynamics. Its error is of order one: the median over checkpoints from step 10^3 on is 0.95–1.69 against 0.32–0.40 for the core, which is the only evidence here that a bounded end-to-end symplectic map retains some of the dynamics at that horizon, a factor of three to four. The final-step error is misleading in 1-D: the reference is periodic with a period that divides 2×10^4 steps (200 periods for `wave1d`, 50 for `wave1d_dir`), so the identity map ends near the initial state and scores 0.15 and 0.19 at the last step, below the core's 1.06 and 0.60. We therefore report the late-window median (Table 13); the 2-D and 3-D problems have no such recurrence. Second, an exactly symplectic core with untrained, fixed gains diverges in all 6 runs (spectral radius 10^3–10^9), so symplecticity is not sufficient. Third, the same core trained for a single epoch is bounded in all 6.

Approximate symplecticity, and what boundedness tracks. Exact symplecticity is not necessary at this horizon either. Perturbing the adjoint's channel mixing by 1% or 10%, which gives

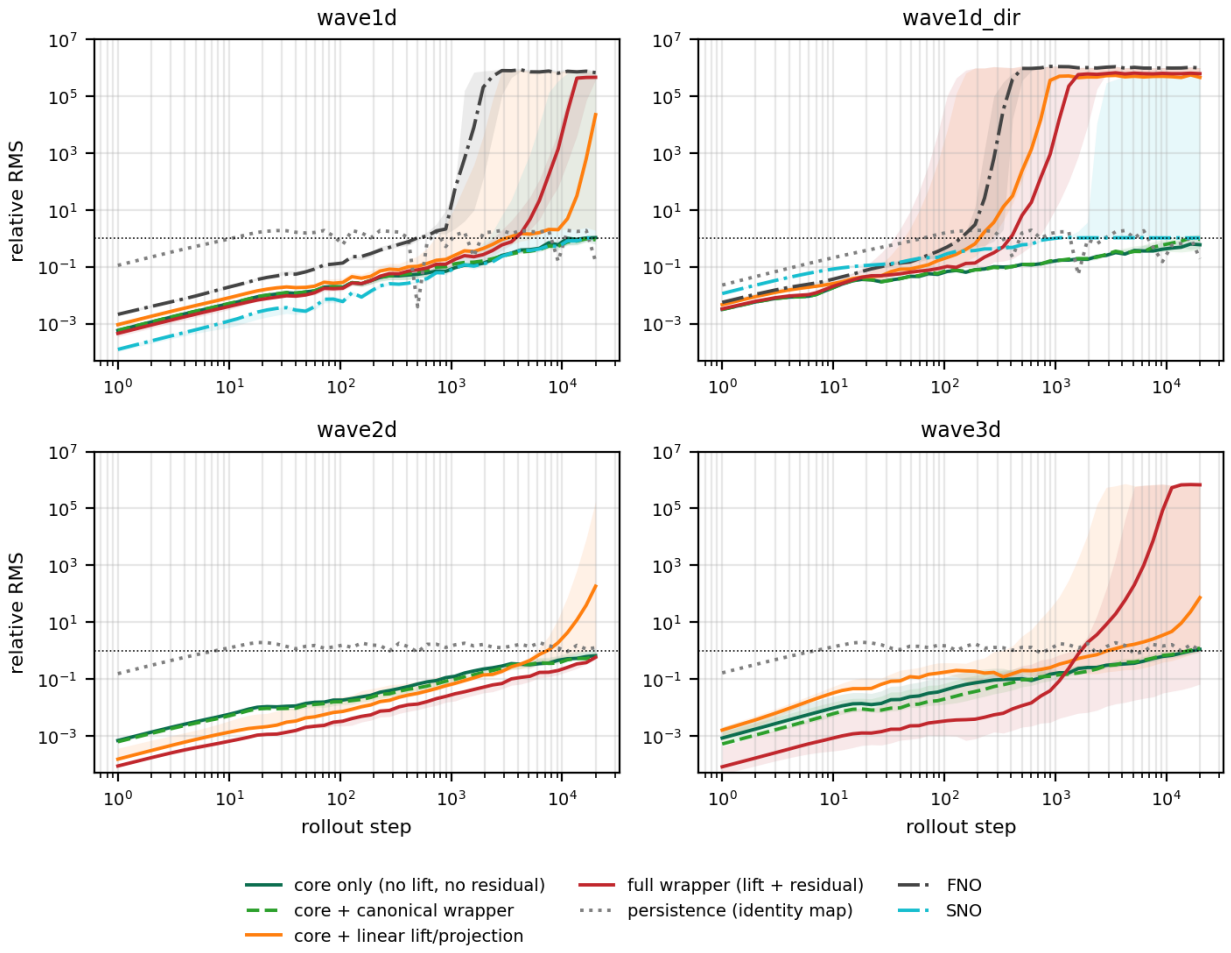


Figure 8: Relative RMS of the staged wrapper ablation on the four wave problems; line: median of three seeds, band: range over seeds. Green solid: the core alone; green dashed: the core in a canonical wrapper; orange: behind a linear lift and projection without residual update; red: the conventional wrapper; grey dotted: the identity map. The dotted horizontal line marks error = signal.

a measured end-to-end defect of 2×10^{-4} to 4×10^{-3}, leaves all 12 runs bounded, with the error at most 1.4× that of the unperturbed core and the energy drift up to 8× higher, while a 100% perturbation (defect 2–3 $\times 10^{-2}$) is unbounded in all 6 (Figure 9b). What boundedness did track is the spectral radius ρ of the one-step Jacobian at the first test state (Figure 9a). Among 102 trained maps on the 1-D and 2-D problems, every one with $\rho \leq 1 + 3 \times 10^{-4}$ was bounded (59 of 59) and one of 43 above it was, an SNO run. That is where linear growth would put the boundary, since $(1 + 3 \times 10^{-4})^{2\times 10^4} \approx e^6$. The core and the canonical wrapper have ρ within 2×10^{-6} of one, with all eigenvalues on the unit circle; the 1% and 10% perturbed maps sit at 1.000005 and 1.00017 (bounded) and the 100% ones at 1.001–1.003 (unbounded); SNO has a median of 1.0006, the linear-lift variant 1.0013, FNO 1.008 and the full wrapper 1.014. A symplectic map need not have $\rho = 1$, as the untrained control shows, but the trained ones did. The end-to-end defect, which the instrument of §3.5 measures, separates the groups less sharply: every map with defect $\leq 10^{-6}$ was bounded (36 of 36), but 22 of 65 above 10^{-3} still were, and across the 114 trained maps its rank correlation with the final error is 0.63. Over training epochs its rank correlation with the growth of the state norm, 0.58, is no higher than that of the one-step validation error, 0.62. The defect says

Table 13: Accuracy of the controls: medians over three seeds at step 2×10^4 (relative RMS, pattern correlation, $|\Delta E/E|$) and of the relative RMS over all checkpoints from step 10^3 (*late*). The 1-D identity map's final-step error reflects an exact recurrence of the reference (see text); its late-window error is the fair comparison.

problem	variant	bounded	rel. RMS, late	rel. RMS, final	pattern corr.	$\lvert\Delta E/E\rvert$
`wave1d`	core only	3/3	0.39	1.06	0.46	2.08
	core + canonical	3/3	0.34	0.89	0.57	2.08
	full wrapper	0/3	3.07	4.6×10^5	0.03	3.0×10^{12}
	persistence	3/3	1.69	0.15	0.98	0
`wave1d_dir`	core only	3/3	0.32	0.60	0.65	6.51
	core + canonical	3/3	0.34	0.92	0.72	17.38
	full wrapper	0/3	6.1×10^5	6.2×10^5	0.02	6.5×10^{13}
	persistence	3/3	0.95	0.19	0.96	0
`wave2d`	core only	3/3	0.34	0.66	0.80	2.14
	core + canonical	3/3	0.33	0.63	0.82	2.17
	full wrapper	3/3	0.12	0.58	0.86	2.66
	persistence	3/3	1.45	1.26	0.16	0
`wave3d`	core only	3/3	0.35	1.11	0.50	1.03
	core + canonical	3/3	0.39	1.22	0.45	1.71
	full wrapper	1/3	132	6.8×10^5	0.01	1.2×10^{12}
	persistence	3/3	1.48	1.01	0.43	0

whether a map is symplectic, not whether it will be stable. Two caveats apply: ρ is evaluated at one state, so it is a local diagnostic rather than a proof, and the correlations pool different families and problems.

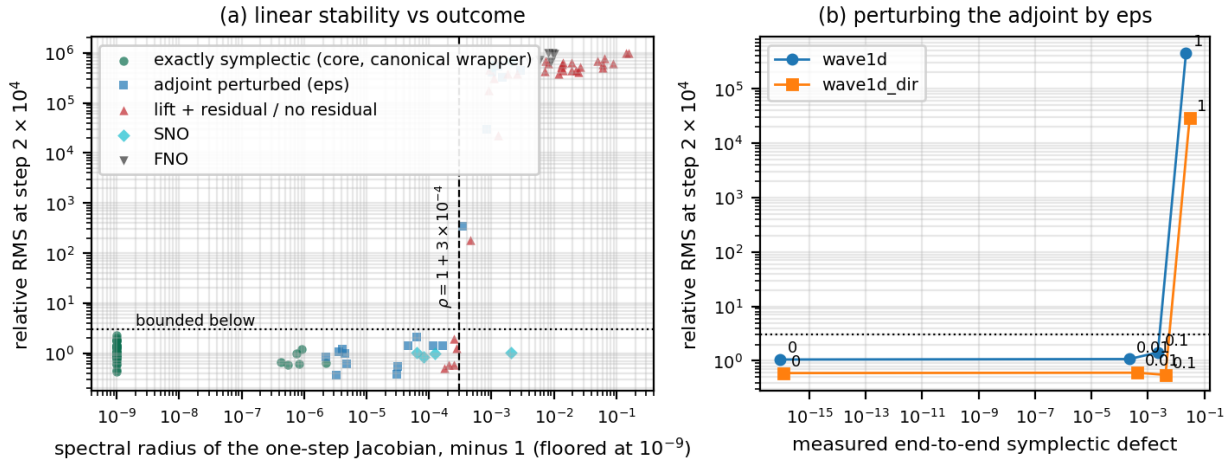


Figure 9: Stability diagnostics. **(a)** Each trained map on the 1-D and 2-D problems: spectral radius of the one-step Jacobian minus one (floored at 10^{-9}) against the relative RMS at step 2×10^4. The dashed line marks $\rho = 1 + 3 \times 10^{-4}$, the dotted line the boundedness threshold. **(b)** Perturbing the adjoint by a relative amount ϵ (labels): end-to-end defect against error, median of three seeds.

Agreement between the two batches. The second batch re-ran the long-horizon runs of the first for the lift-free W_{unif} and lifted W_{unif} families on the four wave problems and for SNO and FNO on the two 1-D problems, with the same seeds (36 paired runs). The classification into bounded and unbounded agrees in 33 of them: the matched-form lift-free operator is bounded in all 12 again, the lifted W_{unif} variant in 4 of 12 again (with the individual seeds on `wave2d` and `wave3d` changing), and SNO in 4 of 6 against 3 of 6.

What this means physically. After 100–200 domain transits the bounded lift-free model still carries a wave of about the right size, travelling with the wrong phase, and it has not conserved energy: $|\Delta E/E|$ on its own trajectory is 1.26–6.25 in the first batch and 1.03–6.51 in the second, an energy error of 103–651%, against at most 0.034% for the reference leap-frog integrator over the same horizon. Since the amplitude is roughly right, the excess must sit mostly in the gradient term $c^2\|\nabla u\|^2$, consistent with the roughening visible in Figures 10 and 11. The best energy behaviour of any bounded trained run in either round, a lifted W_{unif} run on `wave3d`, is 4.3% (second batch) and 6.8% (first batch), still two orders of magnitude worse than the reference. The identity map has exactly zero drift, for the trivial reason that it does not move. No learned operator here conserves energy in a physically useful sense at this horizon, and we do not claim one does.

What it does not mean. Backward-error analysis explains why symplectic *integrators* conserve a modified Hamiltonian over exponentially long times (Hairer et al., 2006), but it assumes a consistent approximation to a specified flow at small step size, and builds a modified equation from exactly those hypotheses. A learned operator satisfies none of them merely by having a symplectic Jacobian: it is not a discretisation of a known Hamiltonian at small Δt, and no modified Hamiltonian is available to appeal to. The energy errors above are what one should expect in the absence of such a guarantee. Makara et al. (2026) prove a modified-Hamiltonian stability result for symplectic learned maps that are smooth Hamiltonian perturbations of the exact flow, under assumptions on the learned error map (a Hamiltonian perturbation representation and a controlled BCH expansion); neither hypothesis is checked for our maps, which at this horizon are far from the exact flow. What we observe is consistent with the qualitative motivation of geometric integration, and we report it on that footing; it is not an instance of the theorem.

Remark 11 (Read the slope with the error). A drift exponent near zero is *not* by itself evidence of health. FNO scores -0.02 on `wave1d` while sitting at relative RMS 7×10^5: once the state has blown up, the drift ratio saturates and its slope becomes meaningless.

Related evidence. Sulskis and Ravi (2026) report a qualitatively similar effect for the metriplectic case: over 200-step rollouts, the unconstrained models they test diverge or collapse while their structure-preserving operator stays bounded, at the cost of accuracy on pure transport and on their smallest backbone. That constructions enforcing different structures (symplectic here, GENERIC there) are both reported to stay bounded where unconstrained models do not suggests the effect is not specific to one bracket; our `wave3d` result shows that exact structure alone does not guarantee it.

The same result seen as fields. Figures 10 and 11 show the state itself rather than summary curves, for the first test trajectory of seed 0, the only seed for which fields were stored. A single trajectory can mislead in both directions, so each row is labelled with the family's population result (median error over seeds and the bounded count). On `wave2d`, seed 0 is the seed in which the lifted W_{unif} variant diverges; in the other two it stays bounded and is more accurate than the lift-free

model. And SNO's plotted trajectory stays bounded on both problems, although averaged over the test trajectories it diverges in most seeds. Read with that caution, the panels show the qualitative failure modes: blow-up (the lifted variants and FNO), a bounded oscillation of the wrong shape (SNO's plotted trajectory on `wave1d_dir`), and gradual roughening at roughly preserved amplitude (lift-free). The bounded-but-wrong case is the one a short-horizon, RMS-only evaluation would pass.

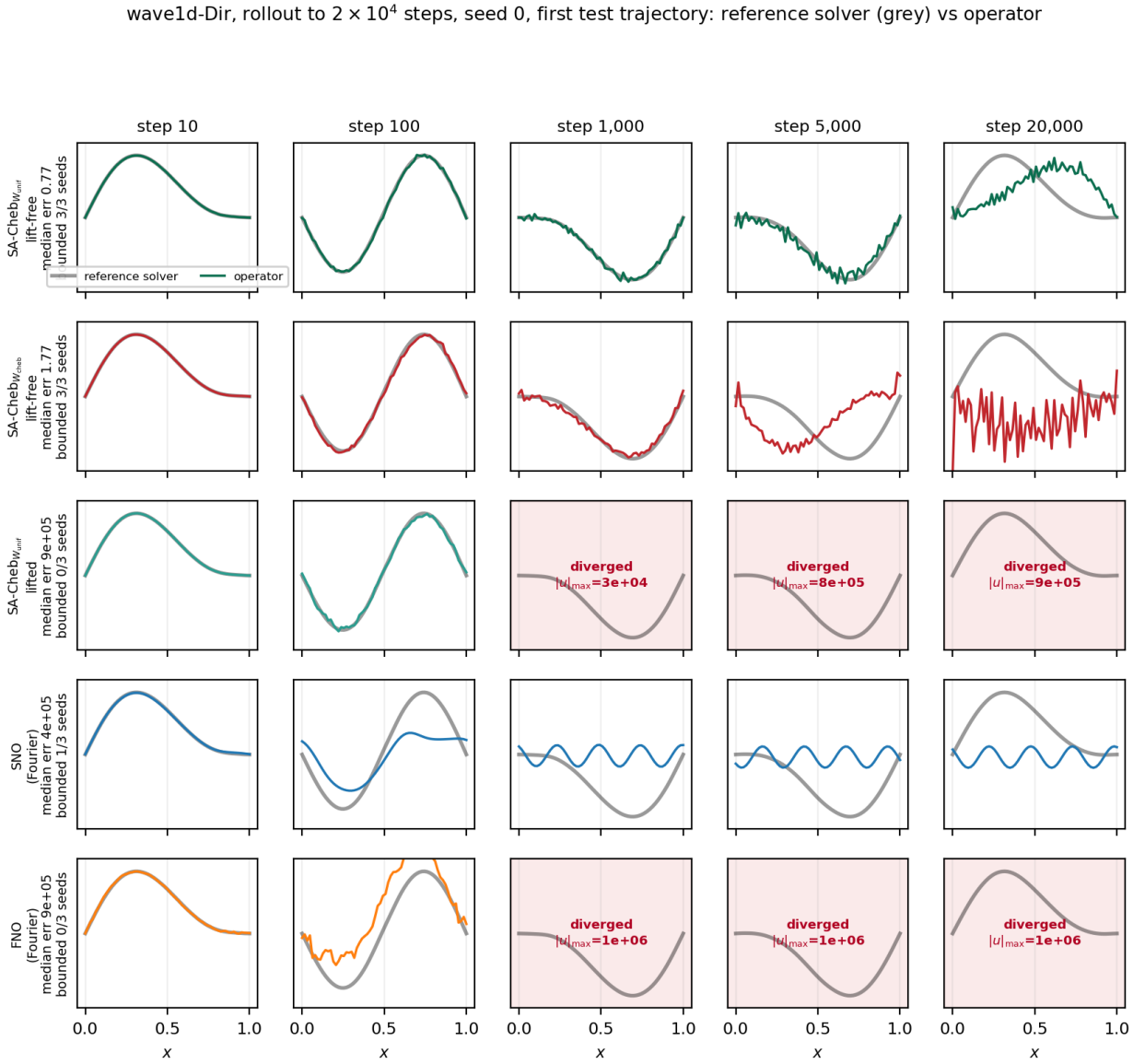


Figure 10: `wave1d_dir`, seed 0, first test trajectory, over the 2×10^4-step rollout: reference solver (grey) against operator prediction. Row labels give each family's population result, median relative RMS at 2×10^4 over seeds and the number of bounded seeds, so a single plotted trajectory cannot be over-read. The lifted SA-Cheb and FNO blow up. SNO's plotted trajectory settles into a bounded oscillation of the wrong shape, though over the test set it diverges in two of three seeds. Both lift-free operators stay bounded with roughly the right amplitude, but by 2×10^4 steps neither tracks the phase (median pattern correlation 0.41 matched, -0.34 mismatched).

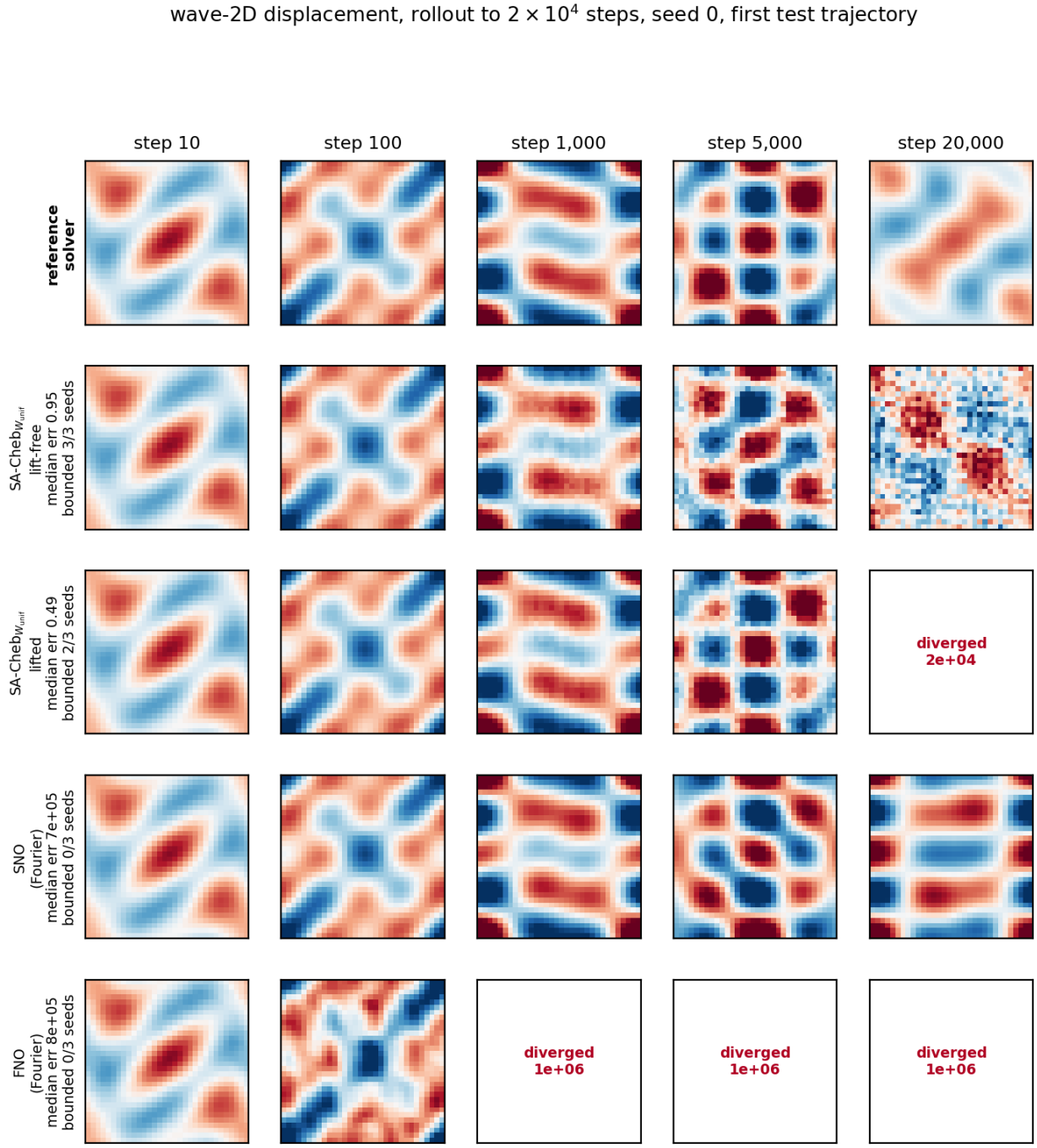


Figure 11: `wave2d` displacement field, seed 0, first test trajectory. The reference solver sustains a coherent interference pattern throughout. The lift-free matched-form operator stays bounded and roughens into noise. In this seed the lifted variant, built from the same gradient-shear construction, has diverged by 2×10^4 and FNO by 10^3; in the other two seeds the lifted variant stays bounded and is the more accurate of the two (Table 11). SNO's plotted trajectory stays coherent, but averaged over the test trajectories its error exceeds 10^4 in every seed. The lifted and lift-free models are matched on parameter *count*, not built from identical tensors: dropping the lift forces $H = C$, so the lift-free model must recover capacity through rank and depth (§4.3).

7 Secondary benchmark

This section places the constructions among standard operators on all nine problems. It is secondary: none of the three research questions rests on it, and Table 14 in particular involves selection over a pool of configurations (§5.3).

7.1 Basis versus boundary

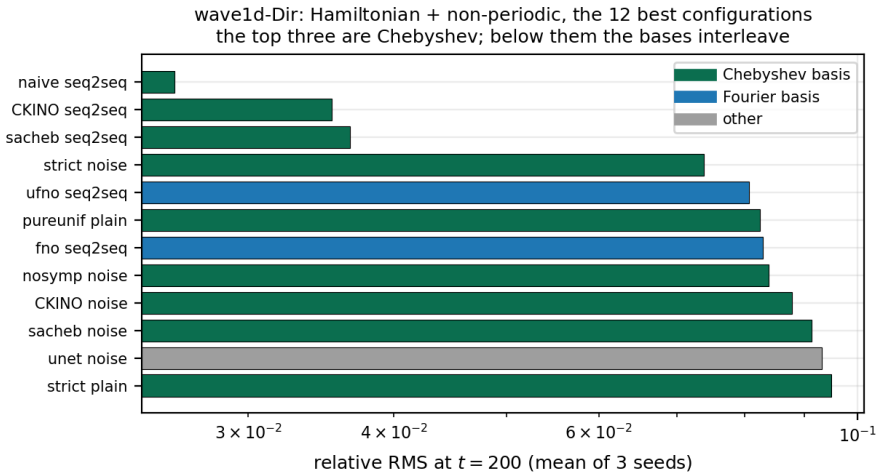


Figure 12: `wave1d_dir`, the Hamiltonian non-periodic problem: the twelve best configurations, mean of 3 seeds. The three best are Chebyshev, led by SA-Cheb (W_{unif}, `seq2seq`), which is 3.1× better than the best Fourier configuration (U-FNO), 3.2× better than the best FNO and 6.5× better than SNO, on a benchmark that follows SNO's boundary and initial-condition protocol. Below the top three the two bases interleave. A plausible mechanism, which we have not isolated, is the Gibbs penalty a periodic basis pays on a non-periodic field (Gottlieb and Shu, 1997).

The pattern extends across the 1-D problems. The Fourier family has the lowest mean error on all five periodic 1-D problems, clearly on two (Burgers, `kdv`) and within seed spread on three (advection, heat, `wave1d`); the Chebyshev family clearly wins the one non-periodic problem. In 1-D the winning basis therefore follows the boundary conditions, with no appeal to structure. That rule does not explain the multi-dimensional problems, which are all periodic: there the Chebyshev family has the lowest mean error on all three, clearly on `wave3d` and `ns2d` and within seed spread of SNO on `wave2d`, and we have no comparable explanation.

7.2 Accuracy across the full matrix

Against the parameter-matched references the margins in higher dimensions are 19.8× over FNO on `wave2d`, 8.8× over FNO and 4.8× over SNO on `wave3d`, and 1.5×/1.4× on `ns2d`. SNO is competitive in 2-D (1.0×).

Table 14 lists a single best model per problem because that is the conventional presentation, but it overstates how decisive most of those rows are. We treat small or seed-sensitive margins as inconclusive rather than applying a formal significance test, because three training seeds are insufficient for reliable inferential statistics. Marking as a tie any problem whose margin between the best two families falls within their combined seed spread (the sum of the two seed standard deviations), only five of the nine have a clear winner (`wave1d_dir`, `wave3d`, `ns2d` to our construction;

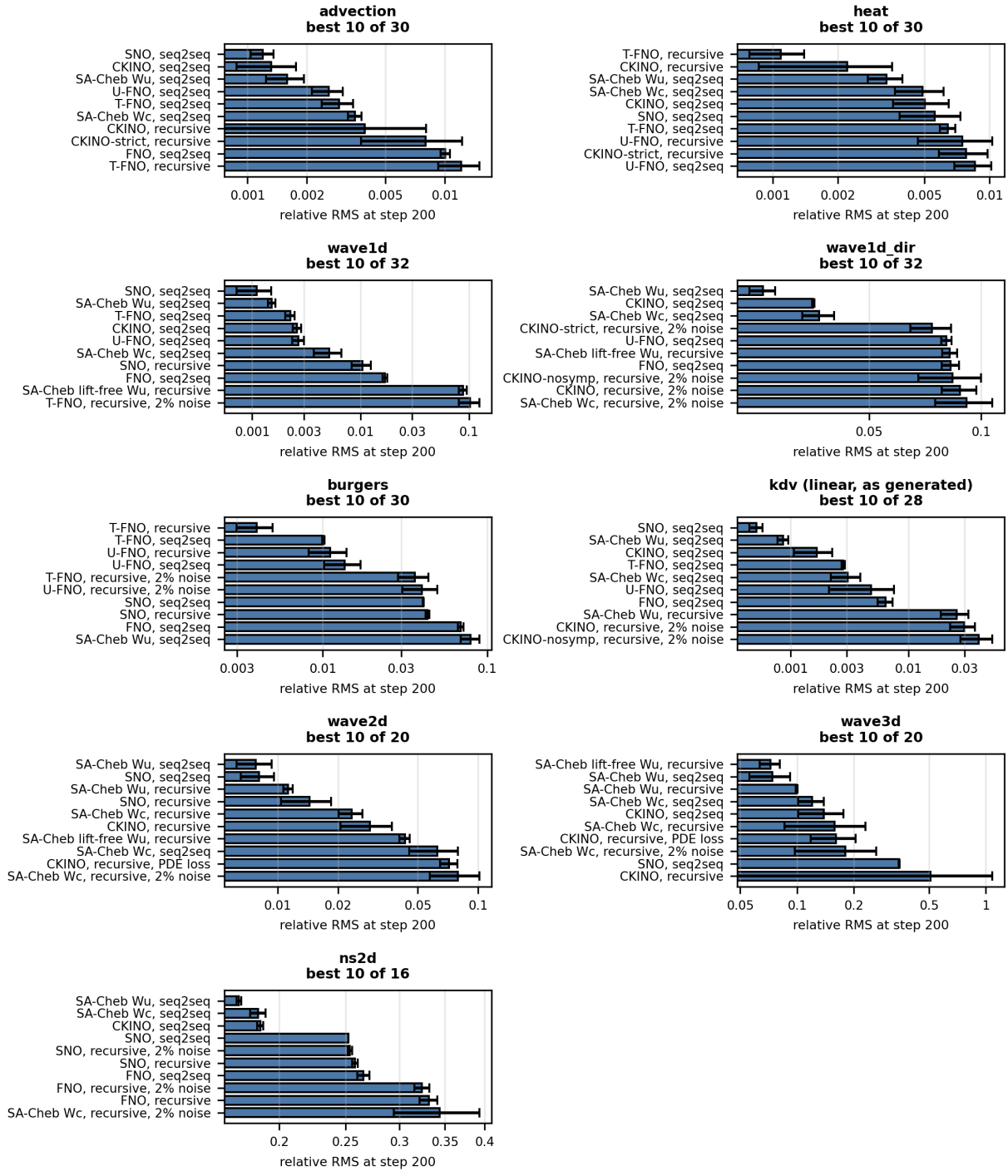


Figure 13: Relative RMS at step 200 (150 in 3-D; log scale, mean ± std over 3 seeds) for the best ten configurations per equation, all matched to ≈25k parameters. Each label gives the model family and the training mode (Table 2; *recursive* with optional input noise or PDE-residual loss, or non-recursive *seq2seq*); W_c and W_u are the Chebyshev and uniform forms. The panel titles give the size of the pool each ten is drawn from; the full listing, with parameter counts and seed spreads, is in the supplementary file `TABLES_MULTISEED.md`. The horizontal scales differ per panel: the spread within a problem ranges from under a decade (`ns2d`) to more than two (`wave1d`). `kdv` is the linear dispersive equation as generated (Table 3).

Table 14: Best configuration per problem: lowest mean over 3 seeds of test relative RMS at step 200 (150 in 3-D), shown as mean ± std. Selected on the test split, so optimistic (§5.3).

problem	best model	relative RMS	basis
advection‡	SNO (seq2seq)	0.001196 ± 0.00016	Fourier
heat	T-FNO (recursive)	0.001090 ± 0.00031	Fourier
`wave1d`	SNO (seq2seq)	0.001103 ± 0.00039	Fourier
`wave1d_dir`	`sacheb_naive` (seq2seq)	0.025939 ± 0.0021	Chebyshev
Burgers	T-FNO (recursive)	0.003963 ± 0.00098	Fourier
`kdv`†	SNO (seq2seq)	0.000511 ± 0.000064	Fourier
`wave2d`	`sacheb_naive` (seq2seq)	0.007771 ± 0.0015	Chebyshev
`wave3d`	`pureunif` (recursive)	0.072375 ± 0.0089	Chebyshev
`ns2d`	`sacheb_naive` (seq2seq)	0.174500 ± 0.0014	Chebyshev

‡ advection is a tie within seed noise: the contending Chebyshev model varies by 15.7× across seeds there, and the winner changed between two executions of the study (§7.4). We do not claim it.
† `kdv` is the linear dispersive equation as generated (Table 3).

`kdv` to SNO; Burgers to T-FNO). Advection, heat, `wave1d` and `wave2d` are ties. We give the resulting tally in §8.

7.3 Unconstrained capacity

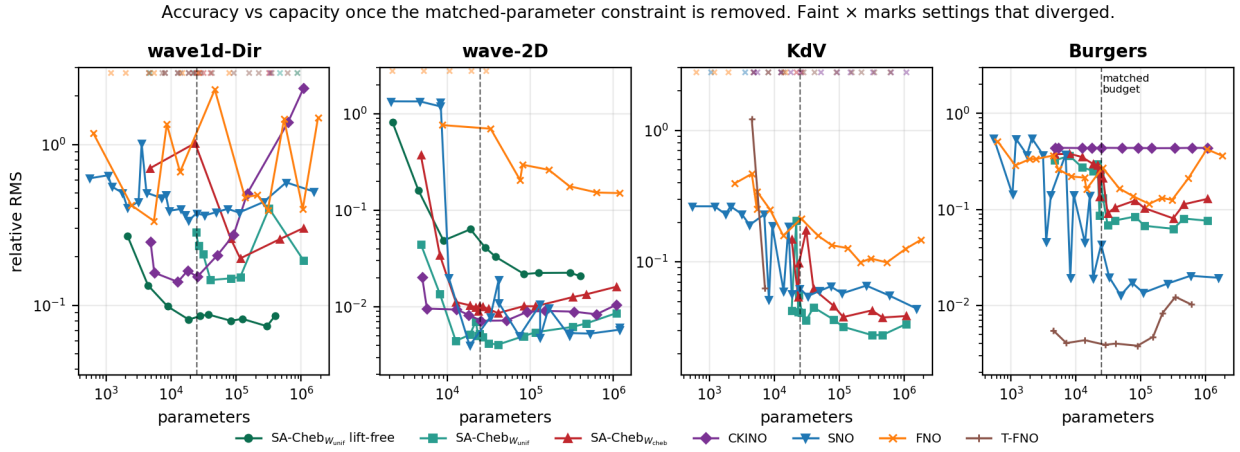


Figure 14: Accuracy against parameter count with the matched budget removed: recursive configurations only, mean of 3 seeds at each width, scored at step 200. The dashed line marks the 25k budget used in Table 14. The ranking is budget-dependent, so the matched table should be read as "best at 25k" rather than "best architecture".

Two observations, both read from seed-averaged results at a fixed width, as plotted; taking each seed's own best width instead would select on noise. On `wave1d_dir` the two end-to-end symplectic models are the best two families in the sweep (0.074 and 0.079), the only problem on which they lead it. And the matched-budget ranking does not survive the removal of the budget: on `wave3d`, SNO fails below ≈ 60k parameters (relative RMS 1.6) but reaches 0.0035 at 262k, seven times better than the best lifted SA-Cheb width (0.025 at 16k), whose error is itself not monotone in width (0.025 to 0.19 between 16k and 38k). Unconstrained, CKINO leads advection by $\approx 10^3\times$ (3.7×10^{-6} at 13k).

7.4 Repeat execution

Operator-learning results are rarely re-run, including by their own authors. Seed counts of three are common, margins of a few percent are reported as wins, and there is usually no way for a reader to tell which of those wins would survive re-running the study; the variance of benchmark comparisons and the practice of reporting it are discussed by Bouthillier et al. (2021), and reproducibility checklists and programmes by Pineau et al. (2021). Part of our matrix exists in two executions on the cluster, which lets us check some of our results. The duplication is partial, so we state exactly what it covers before using it.

What was repeated. Execution B re-ran 657 of the 783 matrix runs and seed 0 of each long-horizon problem. The other 126 matrix runs, namely the lift-free pair on every problem and the SA-Cheb and SNO configurations on `wave2d`, `wave3d` and `ns2d`, exist in a single execution and appear unchanged in both result sets, as do long-horizon seeds 1 and 2. One configuration changed between the executions: on the 1-D problems execution A had placed SNO off the parameter budget (18.5k parameters for the recursive variant, 31k for `seq2seq`), and execution B re-matched it to $\approx$ 25k; every other configuration is identical. Both executions used the same code, seeds and GPU type, on different hosts. T-FNO, DeepONet, the transformer and GENERIC-FNO reproduced bit for bit, FNO in 93 of 102 runs, and every re-run long-horizon SA-Cheb and SNO run did too. The re-run matrix SA-Cheb runs did not: at the same seed they differ from execution A by a median of 13–20% in final test error. We have not isolated the source, which we presume to be floating-point nondeterminism amplified by training. Execution B's per-problem winners were re-selected from its own runs, by the same metric as execution A's (§5.3).

The additional runs of §5.5 are a third, partial check, on a different software version (PyTorch 2.13 against 2.8). They add seeds 3 and 4 for the headline pairs of the matrix, so the lift-free pair, which exists in one execution of three seeds, has five; and they repeat the long-horizon runs of the lift-free and lifted W_{unif} families (and SNO and FNO in 1-D) on all three seeds. Matrix seeds 3 and 4 extend rather than repeat the first three, but they were run independently of them.

We call this a repeat execution rather than a replication, and the distinction is worth keeping. The executions share an implementation, an experimental design, a cluster and a set of authors. The comparison bounds how much of the study is an artefact of one particular execution. It says nothing about whether an independent group writing an independent implementation would reach the same conclusions, which is what replication properly means and is not something authors can certify for themselves.

What survived. Every claim that was re-executed survived at the level at which we make it (Table 15); the per-problem winner on advection did not. There the contending Chebyshev configuration varies by 15.7× across seeds in execution B (0.00062 to 0.00968) against SNO's 1.4×. That is not a close race between two good models; it is one unstable configuration occasionally landing well, and under execution A alone it would have been reported as a win for our own operator. Below the level of the claims the noise is larger still: the recursive heat pair of the form ablation favoured the matched form by 1.58× in execution A and the mismatched form by 1.02× in execution B, with the same seeds. Which problems count as clear wins also moved: heat was clear for T-FNO in A and a tie in B, `kdv` a tie in A and clear for SNO in B, though our construction's share of the tally did not change.

What follows. First, a working rule rather than a statistical test: with three seeds we treat a margin that falls within the combined seed spread of the two contenders as inconclusive, not as

Table 15: Headline claims under repetition. The first result column gives the main-matrix result; the second the repeat or extension in the additional runs (PyTorch 2.13). *Not repeated* marks claims that rest on one execution.

claim	main matrix	additional runs	status
form ablation, lifted	16 of 18 pairs	17 matched, 5 inconclusive, 0 other (of 22)	reproduced
lift-free pair, ratios	17.2/1.24/21.6/14.2×	17.3/1.07/21.3/14.4× (five seeds)	extended, agrees
long-horizon, lift-free W_{unif}	12 of 12 bounded	12 of 12	reproduced
long-horizon, lifted W_{unif}	4 of 12	4 of 12; 2 runs changed class	reproduced in count
long-horizon, SNO (1-D)	3 of 6	4 of 6	one run changed
long-horizon, mismatched lift-free	9 of 12	not repeated	not repeated
winner, 1-D, 5 of 6 problems	unchanged between executions A and B		stable
winner, advection	CKINO 0.001092	SNO 0.001196 (execution B)	flipped
tie tally, ours / SNO / T-FNO	5.0/2.0/2.0 (A)	5.0/2.5/1.5 (B)	ours stable

a result (§7.2); with five seeds we apply an interval and an effect floor instead (§5.5). The re-run shows why, since re-running alone moved the same configuration by 13–20%. Second, a reporting standard: the seed spread should be published alongside the mean, because it is what tells a reader whether a margin is load-bearing, and we report both throughout. Third, the claims marked *not repeated* in Table 15 rest on one execution, and the winners of the 2-D and 3-D problems in Table 14 still come from one.

7.5 Supporting axes

Darcy (elliptic, Dirichlet). The problem is $-\nabla\cdot(a\nabla u) = 1$ on $(0,1)^2$ with $u = 0$ on the boundary and a smooth log-normal permeability a drawn from a random field with 8×8 modes, solved by a five-point finite-difference scheme and conjugate gradients, with 512 training samples at 64^2. CKINO attains 8.7× lower relative L^2 error than FNO overall and 18.3× lower *at the boundary*, transferring zero-shot from 64^2 to 128^2. This is the clearest non-periodic win for the Chebyshev basis.

Capacity ceilings. A 6k–400k sweep shows CKINO's Burgers failure is *representational*, flat at ≈ 0.25 across a 60× range, and its recursive instability on `kdv` persists at every budget. Neither is fixed by scale.

Discretisation invariance. Training at $N = 64$ and evaluating at $N = 128$, **SNO and T-FNO win this axis outright**: their error changes by less than 2% between the two grids *and* they are accurate, which is the combination that matters. CKINO with a convolutional lift degrades by 57× on advection. The spectral lift restores invariance (0.998) but trains unreliably at the training grid: one seed matches the convolutional lift (1.2×10^{-3}) and two fail (0.59, 0.84). There is no setting of our construction that is both invariant and reliably accurate, and we do not have a fix. If a workflow requires evaluating at a resolution it did not train on, that is a decisive argument for the Fourier family and against ours.

Unseen resolutions. Training on two resolutions and testing the one-step error on four unseen ones (§5.5) separates the bases at the coarse end only (Figure 15). At the coarsest unseen resolution (25 CGL nodes or 24 points) the Chebyshev operators' error is 5–19× (CGL) and 11–34× (Dirichlet) that of the best Fourier operator, SNO or FNO. At the finer unseen resolutions all models are within 1.2–1.5× of each other. This is the same verdict as for the 64 → 128 test above and extends it: the Chebyshev construction does not transfer to coarser grids, and the Fourier operators do.

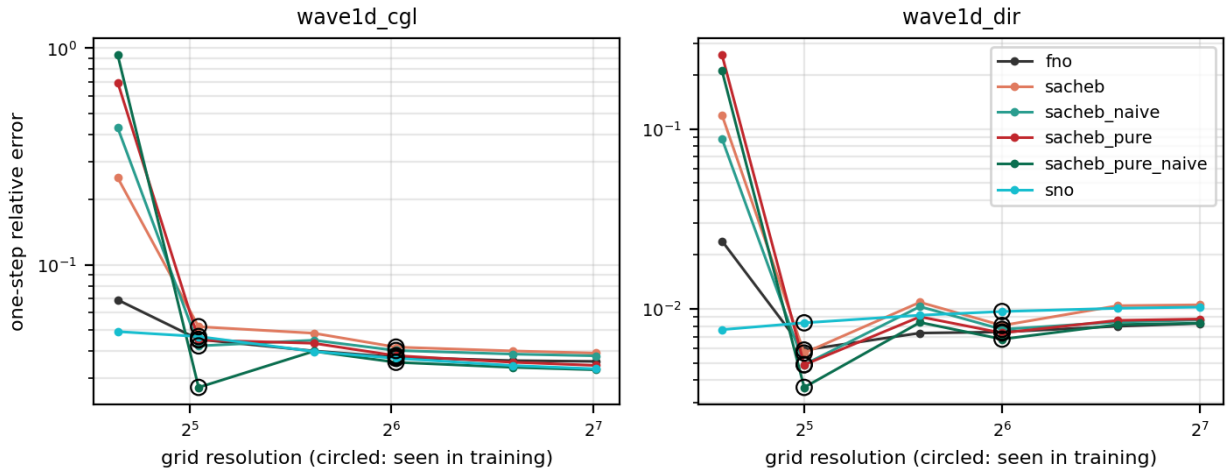


Figure 15: One-step relative error on resolutions seen (circled) and unseen in training, geometric mean over three seeds. At the coarsest unseen resolution the Chebyshev operators (`sacheb` variants) are one to two orders of magnitude worse than SNO and FNO; at finer resolutions the models are within about 1.5×.

Inference speed. Operators beat the reference solver only where the solver is expensive: FNO is 3.6× faster on Burgers, 8.9× on `kdv` and 5.6× on `ns2d`; CKINO is 2.6× faster on `kdv` but 0.07× on advection. These are single wall-clock measurements and moved by up to 15% between the two executions.

8 Where each construction wins

We want to be explicit about the shape of this table rather than let it be read selectively. Treating a margin that falls within the combined seed spread of the two best families as inconclusive (§7.2), only *five* of the nine problems have a clear winner: `wave1d_dir`, `wave3d` and `ns2d` to our construction, `kdv` to SNO, Burgers to T-FNO. The remaining four (advection, heat, `wave1d`, `wave2d`) are ties. Splitting ties evenly the tally is ours 5.0, SNO 2.5, T-FNO 1.5; in execution A it was 5.0, 2.0, 2.0 (§7.4).

Our construction wins the two axes it was designed for, with qualifications: reliability under long rollout, where the matched-form end-to-end symplectic variant never diverged but is not accurate at that horizon (and an identity map is bounded too), and accuracy on the non-periodic problem and the multi-dimensional ones, clearly on three of four. It loses the two it was not designed for. **SNO wins discretisation invariance outright**, being simultaneously invariant and accurate where our operator can be reliably only one or the other, and the Fourier family has the lowest mean error on every periodic 1-D problem. A reader whose workload is periodic, or who must evaluate at a resolution the model was not trained on, should use a Fourier operator.

Table 16: Where each construction wins and loses. No single operator dominates; the axes separate, and two of the four axes go to the Fourier family.

construction	wins	loses
`pureunif` (lift-free, W_{unif})	long-horizon reliability (the only family bounded in all 12 runs at 2×10^4 steps); `wave3d` at 25k; top of the unconstrained `wave1d_dir` sweep	bounded but not accurate at long horizon; needs a canonical (q, p) pair, so undefined on scalar fields; limited capacity per parameter
`sacheb_naive` (lifted, W_{unif})	`wave1d_dir` and `ns2d` at 25k, `wave2d` tied with SNO; the most accurate long-horizon runs on `wave2d`/`wave3d` when it stays bounded	no end-to-end guarantee; bounded in only 4 of 12 long rollouts; 5–34× worse than the Fourier operators at the coarsest unseen resolution
CKINO	advection unconstrained by $\approx 10^3\times$; Darcy by 8.7× (18.3× at the boundary)	not symplectic in any form; Burgers ceiling; instability on `kdv`; worst discretisation invariance in the study
SNO (Makara et al., 2026)	`kdv` outright, and ties advection, `wave1d` and `wave2d`; **discretisation invariance outright**; best unconstrained `wave3d` result	Fourier basis mismatched on non-periodic domains (6.5× on `wave1d_dir`); fails on `wave3d` below $\approx$ 60k parameters
T-FNO (Kossaifi et al., 2023)	Burgers outright, heat tied with CKINO; invariance	no structure; diverges in 5 of 6 long rollouts

9 Limitations

The form is not separated from the quadrature rule. In the implementation the weight W defines the quadrature of the kernel integral inside K as well as the adjoint K^* (§4.2), so the matched and mismatched variants differ in the forward operator too. The experiments of RQ2 therefore show that using the grid's quadrature weights throughout helps where the grid has a standard quadrature; they do not show that the *symplectic form* is what matters. The control that would separate the two keeps the kernel quadrature fixed and varies only the weight in the adjoint. It is straightforward to build and has not been run, and until it is the title question is answered for the operator as a whole rather than for the form alone.

The matching hypothesis is only partly supported. Both directions are tested, on three 1-D grids, with five seeds. The preference follows the grid on uniform grids (17 of 22 comparisons) and on CGL nodes (two of three), but the non-recursive CGL pair is a tie and the mapped grid, where neither standard form is the quadrature, shows no benefit from the exactly matched weight. The two non-uniform grids are 1-D and Dirichlet, and the reversal is much smaller (1.4×) than the forward effect (14–21×); we have not explained why.

The batches differ, and repetition is partial. The additional runs used a different software version (PyTorch 2.13 against 2.8) and 32 test trajectories per matrix run against 12, from the same split seeds, and we pool seeds 0–2 with 3–4 only through the per-seed error. 126 of the 783 main-matrix runs, and the mismatched lift-free long-horizon runs, exist in a single execution (§7.4).

Winner selection in Table 14. The best configuration per problem is chosen on the test split rather than on validation, so those per-problem numbers are optimistic by an unquantified amount (§5.3). The paired comparisons the argument rests on are not affected, since no configuration is

selected from a pool.

The wrapper ablation is confounded with capacity. The staged ablation of §6.3 shows that the non-symplectic linear lift and projection, not the residual update, is where boundedness is lost, but the canonical wrapper adds three parameters while the lifted variants have a different width and rank at the same budget. "Symplectic wrapper versus not" is therefore crossed with the size of the wrapper, which we did not vary independently. A canonical wrapper with a comparable number of parameters would remove the confound.

The controls are small and the diagnostics local. The spectral radius that tracked boundedness is evaluated at one state, for the 1-D and 2-D problems only, and the correlations of §6.3 pool families and problems. The identity map shows that boundedness alone is weak evidence, and the evidence that the bounded symplectic maps keep some dynamics is a factor of three to four in late-window error over that map.

Fields are shown for one seed. Field snapshots were stored for seed 0 only, so Figures 10 and 11 show one trajectory of one seed; the population numbers are in Tables 10, 11 and 12.

Parameter count is the only matched resource. Equal parameters does not mean equal FLOPs, memory or wall-clock across families with arithmetic profiles as different as these (§5.4); accuracy comparisons here should not be read as efficiency comparisons.

Re-implementations. SNO, GENERIC-FNO, FNO, T-FNO, U-FNO, U-Net, DeepONet and the Transformer are our own implementations from the published descriptions, not the authors' or library code, and were not tuned to the authors' protocols. In particular SNO is run here at 25k to 262k parameters, whereas its authors train models of about 2.3M parameters on 10^5 one-step pairs (Makara et al., 2026); our SNO numbers are not a reproduction of theirs.

Our GENERIC-FNO implementation. It diverged in all 12 entries. Parameterising the PSD multiplier as $M = b^2$ with b initialised at zero places it at a saddle ($\partial M/\partial b = 2b = 0$), so the dissipative channel never activates. This is a defect of our implementation, *not* evidence about the published method, and no conclusion should be drawn from those rows.

The `kdv` problem is linear as generated. The exponential integrator that generates `kdv` multiplies its nonlinear term by an extra factor of the micro-step, so the stored trajectories are those of the linear dispersive equation $u_t = u_{xxx}$ (Table 3). Every result labelled `kdv` is a result on a linear problem, and the benchmark contains no nonlinear dispersive case: its nonlinear coverage rests on Burgers and `ns2d` alone. A correctly written solver (problem `kdv_nl` of the accompanying code) conserves $\int (u_x^2 + 2u^3)\,dx$ for $u_t + 6uu_x - u_{xxx} = 0$ to 4×10^{-5} over 100 stored steps, and the benchmark should be re-run with it before any claim about solitons is made. The Burgers reference is first-order in time (ETD1), at a micro-step of $\Delta t/8$, and the other reference solvers are as in Table 3.

Canonical structure only. The lift-free construction needs a canonical (q, p) pair and is therefore undefined on the scalar-field problems (advection, heat, Burgers, `kdv`, `ns2d`). The Korteweg–de Vries equation is Hamiltonian only under the non-canonical bracket of Gardner (1971), which we do not cover.

Scale and horizon. Domains are small (32–64 points in 1-D, 32^2, 16^3, 64^2), adequate to resolve the reported orderings but not a deployment benchmark. 2×10^4 steps separates bounded from unbounded but is still far short of the regimes where backward-error analysis is usually invoked, and the capacity sweep of Figure 14 covers recursive configurations only.

Single hardware. All runs of both batches used one accelerator type (Tesla T4) with TF32 disabled, so arithmetic is consistent across the matrix; results may differ on accelerators with different defaults.

Availability. Per-run result files, the analysis scripts and the figure generators are archived with the project code. Model checkpoints and stored field snapshots (seed 0 only) are not archived,

so the field figures cannot be regenerated from archived data alone.

10 Conclusion

We set out to test whether geometric structure helps a learned operator and found that the question, as usually posed, is incomplete. Symplecticity is defined relative to an inner product; on a non-uniform grid two reasonable implementations preserve two different forms, both exactly, and only measurement distinguishes them. Once the form is named the picture is clearer but not simple. Matching the weights to the grid helps where the grid has a standard quadrature, in both directions: on uniform grids in 17 of 22 comparisons and by 14–21$\times$ for the end-to-end symplectic pair, and on CGL nodes in two of three, by a smaller margin. It brings nothing on a mapped grid where neither standard form is right, and because the weights also define the kernel's quadrature, it is not yet attributable to the symplectic form rather than to the quadrature rule; the control that would settle that is the next experiment. Symplecticity that holds end to end went with bounded rollouts of 2×10^4 steps in 60 of 63 runs, and a non-symplectic linear lift and projection lost it in most, but the controls limit what that shows: an identity map is bounded, an untrained symplectic map is not, a small perturbation of the adjoint leaves the map bounded, and what boundedness tracked was the spectral radius of the one-step Jacobian. The bounded maps are not accurate at that horizon, with errors comparable to the signal and energy off by more than 100%. In these experiments, then, structure preservation buys reliability rather than accuracy, and the relevant property looks more like near-neutral linear stability than exact symplecticity. We also show by example that a unit Jacobian determinant is not preservation of ω, and we think the general lesson is that such claims should be measured rather than argued: at scale the measurement needs only standard Jacobian–vector and vector–Jacobian products and does not form the Jacobian.

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A A volume-preserving update that is not symplectic

Each Störmer–Verlet-style half-step of the CKINO block (§4.1) has a Jacobian that is triangular with unit diagonal blocks, hence determinant 1. It is tempting to conclude that each half-step preserves ω, but the step from "determinant 1" to "preserves ω" is invalid. By Proposition 2, $\det D\Phi = 1$ holds for every shear, while preservation of ω_W additionally requires $WA = A^\top W$. The kernel (12) of CKINO uses independent φ_r, ψ_r and an unconstrained channel mix, so A is not self-adjoint, and the measured defect is ≈ 1.4 in *both* candidate forms (Figure 5). The same step bears on the energy behaviour of the block: the backward-error analysis of symplectic integrators, which gives a modified Hamiltonian conserved over exponentially long times (Hairer et al., 2006), requires symplecticity, and volume preservation alone does not imply the existence of a modified Hamiltonian.

Scoring against a single inner product. A defect measured against one form only can mislabel a construction. The uniform-form variant of SA-Cheb has an $O(1)$ defect when scored against W_{cheb} alone and would be called non-symplectic; scored against both forms (Figure 5) it is exactly symplectic in W_{unif}. This is why the defect is always reported against both inner products.

B The weighted adjoint in the low-rank basis

With $\alpha_r = \langle \psi_r, v\rangle_W$ and $(Kv)(x) = \sum_r \varphi_r(x) M_r \alpha_r$,

$$\langle Kv, g\rangle_W = \sum_r (M_r\alpha_r)^\top \langle \varphi_r, g\rangle_W = \sum_r \alpha_r^\top \left(M_r^\top \langle \varphi_r, g\rangle_W\right) = \Big\langle v,\ \sum_r \psi_r\, M_r^\top \langle \varphi_r, g\rangle_W \Big\rangle_W,$$

which is exactly $\langle v, K^* g\rangle_W$. Hence the weighted adjoint is obtained by exchanging φ and ψ and transposing the channel mixing; no inverse of W is formed and the cost remains $O(RN^d)$. In d dimensions the weight is the outer product of the per-axis weights and remains diagonal, so both this identity and Theorem 5 carry over unchanged.